Classical training of conditional instantaneous quantum polynomial-time Born machines

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Instantaneous quantum polynomial-time (IQP) Born machines are attractive quantum generative models, as some instances are conjectured to be hard to sample classically, yet they can be trained classically. Standard IQP Born machines apply a learned IQP unitary to a fixed input state to represent a single distribution. Conditioning them requires either fitting condition-dependent parameters in the diagonal layer, such that the condition is absorbed into the learned parameters, or adding external control to the circuit. In this work, we introduce conditional IQP Born machines, in which the condition is instead encoded in the input state and a single IQP unitary is shared across conditions to model a family of distributions. We show that efficient sampling and amplitude-ratio evaluation for the state entering the diagonal layer suffice to obtain unbiased classical estimates of a conditional maximum mean discrepancy objective and its gradients. Because this requirement concerns classical access rather than gate structure, the encoder need not have IQP form, which allows us to encode conditions in a more general manner and to enforce output bit-flip symmetries without restricting the learned unitary. We also give a sufficient condition under which an encoder reduces to condition-dependent parameters of the diagonal layer. In sixteen-qubit experiments, we show that conditional IQP Born machines can jointly control the mean and width of Hamming-weight distributions and generalize to condition combinations not seen during training. Conditioned on the inverse temperature β, they also capture the unimodal-to-bimodal magnetization transition of the Curie–Weiss model at held-out temperatures. For this model, an encoder whose reference states are not IQP states roughly halves the full-distribution error of an encoder whose reference states are IQP states. These results extend IQP Born machines to conditional generation while retaining classical trainability, expanding the potential applications of IQP circuits.

I. INTRODUCTION

Conditional generative models approximate a family of target distributions $p(\cdot \mid c)$ with a single model. They capture how a distribution depends on one or more conditioning variables, such as the temperature of a physical system or a class label in image generation. Learning across conditions aims to capture structure shared by the family, which in turn supports generation at conditions not encountered during training. In classical machine learning, examples include conditional variational autoencoders [1–3], generative adversarial networks [4–6], and diffusion models [7, 8]. Among quantum generative models, a widely studied architecture is the quantum circuit Born machine (QCBM), which represents a probability distribution through the measurement statistics of a parameterized circuit [9, 10]. QCBMs have been extended to conditional generation through input-state preparation, conditional registers, and postselection [11–15]. Kiss *et al.*, for example, map each condition to an input state through a condition-dependent feature map and train a single circuit shared across conditions [12]. Training such models, however, generally requires circuit samples, obtained either from quantum hardware or from classical simulation whose cost grows exponentially with the number of qubits. On hardware, gradient estimation also lacks the scaling of backpropagation unless the circuit has special structure, such as commuting generators [16].

Instantaneous quantum polynomial-time (IQP) circuits [17] offer a particularly attractive design for QCBMs [18], as certain families are conjectured to be classically hard to sample from under standard complexity-theoretic assumptions [19–22]. Nevertheless, classical algorithms can efficiently estimate their Pauli-Z expectation values to additive accuracy [23]. This distinction enables classical training of IQP Born machines with objectives that can be expressed through these expectation values [24]. The maximum mean discrepancy (MMD) with a Gaussian kernel is one such objective, as it can be written as a weighted average of squared differences between model and target Pauli-Z expectation values [25, 26]. We call a model *classically trainable* when its training objective and gradients can be efficiently estimated to a prescribed additive accuracy on a classical computer. Recio-Armengol *et al.* combined these ingredients into a framework in which IQP Born machines are trained classically and require quantum hardware only for sampling [27], an approach since applied to calorimeter images and extended to other classically trainable architectures [28, 29].

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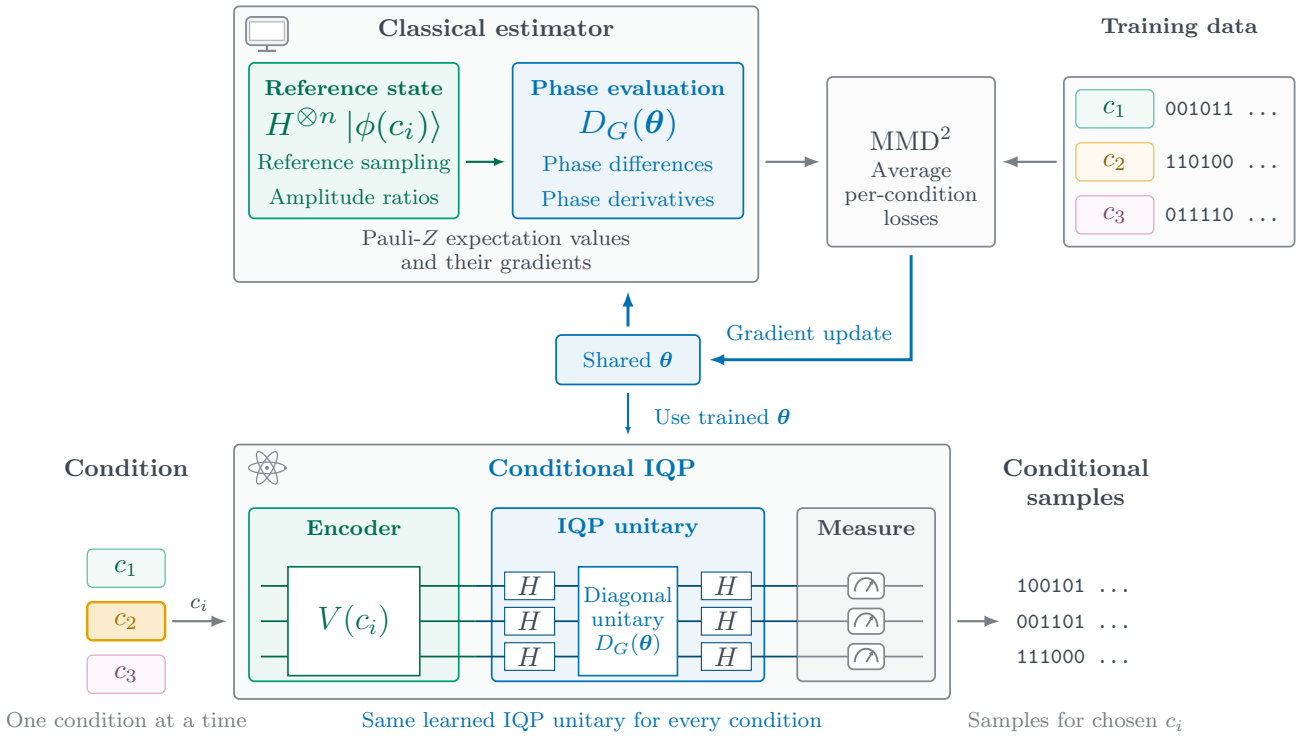


FIG. 1. Proposed framework for classical training of conditional instantaneous quantum polynomial-time (IQP) Born machines. An encoder maps each condition c_i to an input state $|\phi(c_i)\rangle$. During classical training, samples and amplitude ratios from the reference state $H^{\otimes n}|\phi(c_i)\rangle$, together with phase differences induced by the diagonal layer, yield estimates of conditional Pauli-Z moments and their gradients. We use these estimates to optimize a single IQP unitary shared across conditions with a squared maximum mean discrepancy (MMD) objective averaged across training conditions. During inference, a chosen condition c_i determines the input state. Applying the trained IQP unitary $U_{\boldsymbol{\theta}}$ to the input state $|\phi(c_i)\rangle$ and measuring in the computational basis produces samples from the learned conditional distribution $q_{\boldsymbol{\theta}}(\cdot \mid c_i)$.

In the standard classically trainable formulation, the IQP unitary acts on the fixed input state $|0\rangle^{\otimes n}$, so each parameter vector represents a single distribution. Conditioning such a model requires either fitting condition-dependent parameters in the diagonal layer or adding external control to the circuit. In the first approach, Liu *et al.* retain a shared circuit, fit a small parameter block separately for selected conditions, and interpolate the fitted parameters to generate samples at unseen conditions [30]. They subsequently used physics-informed initialization to improve this condition-specific adaptation [31]. In the second approach, ConQuER fits an additional controller to data filtered by a target condition while keeping a pretrained generator fixed [32]. In both approaches, the dependence on the condition is fitted separately for individual conditions rather than learned as a function of the condition. Generation at a new condition therefore requires a new fit, a new filtered dataset, or interpolation between fitted parameters.

In this work, we instead encode the condition in the input state of a classically trainable IQP Born machine. An encoder prepares a condition-dependent state $|\phi(c)\rangle$, and a single IQP unitary $U_{\boldsymbol{\theta}}$, shared across conditions, defines $q_{\boldsymbol{\theta}}(x \mid c) = |\langle x|U_{\boldsymbol{\theta}}|\phi(c)\rangle|^2$. Training this unitary jointly across the family allows the model to learn the dependence on the condition as a function, rather than fitting it separately at each condition. Input conditioning itself has precedent in QCBMs [12]. Our setting differs in two respects: training remains classical, and the encoder need not have IQP form. Figure 1 summarizes the framework.

We extend the framework of Recio-Armengol *et al.* from a fixed input state to condition-dependent input states [27]. Classical training requires only efficient sampling and amplitude-ratio evaluation for the reference state $H^{\otimes n}|\phi(c)\rangle$. Under this access, we adapt established sparse-observable estimators [23, 33] to obtain unbiased estimates of a conditional MMD objective and its gradients. Because this requirement constrains classical access rather than gate structure, the encoder need not have IQP form and may prepare correlated input states. We use this freedom in two ways. First, we encode conditions in correlated input states, including one whose reference states lie outside the class of IQP states. We show that encoders which only apply condition-dependent phases cannot be absorbed into the learned parameters at the

level of output states, yet their reference states are themselves IQP states. Second, we construct condition-dependent input states whose correlations enforce prescribed output bit-flip symmetries for every condition and parameter vector. That work used GHZ input states to impose global bit-flip symmetry on an unconditional model and outlined extensions to other symmetries [27]. Our constructions carry such symmetries over to conditional models, and Appendix D gives an example with six independent switching symmetries.

We test the framework numerically at sixteen qubits, where exact output probabilities are available. We first train a conditional IQP model to shift the mean of binomial Hamming-weight distributions across a continuous condition interval. In a two-coordinate extension, we train the model to control the mean and width of Hamming-weight distributions jointly. We train it only on conditions that vary one coordinate at a time and evaluate it at sixteen held-out combinations of both coordinates, twelve of which lie outside the convex hull of the training conditions. At these combinations, the mean full-distribution total variation distance (TVD) is 0.047. For the Curie–Weiss model, we condition a single model on inverse temperature β and find that it captures the transition from a unimodal magnetization profile in the hot regime to a bimodal profile in the cold regime at held-out temperatures. A squeezed parity encoder, whose reference states are not IQP states, reduces the temperature-averaged full-distribution TVD from 0.103 with a phase encoder to 0.047 under the same protocol, and it brings the energy-fluctuation response close to the target. With the same encoder, the shared model comes close to circuits trained separately at each test temperature, with mean full-distribution TVD 0.053 over the three test temperatures against 0.050. Finally, we classically train conditional models with up to 2048 qubits and observe decreasing estimated MMD at held-out conditions, which establishes computational feasibility rather than full-distribution accuracy at this scale.

These results indicate that a single classically trained IQP unitary can represent a family of distributions and generalize across conditions. The target families studied here admit efficient classical sampling, so our experiments do not address quantum sampling advantage. Section II introduces the conditional model, its classical estimators and input-state-induced symmetries. Section III presents the numerical experiments, Sec. IV discusses limitations and implications, and Sec. V concludes.

II. CONDITIONAL IQP BORN MACHINES

A. Model and reference-state access

The conditional model separates condition encoding from the learned unitary. Consider a target family of distributions $\{p(\cdot \mid c) : c \in \mathcal{C}\}$ over n-bit strings. An encoder maps each condition c to an input state $|\phi(c)\rangle$, and we apply the same parameterized unitary $U_{\boldsymbol{\theta}}$ to every encoded state. The model distribution is

$$q_{\boldsymbol{\theta}}(x \mid c) = \left|\langle x| U_{\boldsymbol{\theta}} |\phi(c)\rangle\right|^2, \qquad x \in \{0,1\}^n. \tag{1}$$

We parameterize $U_{\boldsymbol{\theta}}$ using an undirected graph $G = (V, E)$ with $n = |V|$ vertices. Each vertex $i \in V$ contributes a one-body generator Z_i, while each edge $\{i, j\} \in E$ contributes a two-body generator $Z_i Z_j$. We write the unitary as

$$U_{\boldsymbol{\theta}} = H^{\otimes n} D_G(\boldsymbol{\theta}) H^{\otimes n}, \tag{2}$$

where

$$D_G(\boldsymbol{\theta}) = \exp\left(i \sum_{i \in V} \theta_i Z_i + i \sum_{\{i,j\} \in E} \theta_{ij} Z_i Z_j\right). \tag{3}$$

Although we restrict the ansatz in our numerical experiments to one- and two-body generators, the formulation extends directly to higher-order interactions by including additional k-body Pauli-Z generators.

Throughout this work, "IQP unitary" refers exclusively to the learned block $U_{\boldsymbol{\theta}}$, and "diagonal layer" to its phase layer $D_G(\boldsymbol{\theta})$. The encoder is not restricted to IQP form and may include non-diagonal operations, so the complete conditional circuit need not belong to the IQP circuit class.

The initial Hadamard layer maps the encoded input to the reference state $|\psi_c\rangle = H^{\otimes n} |\phi(c)\rangle$, immediately before the diagonal layer. We assume that each encoded input is normalized and independent of the learned parameters $\boldsymbol{\theta}$, so the reference state inherits the same properties. We train the model using samples and amplitude ratios from this reference state. Its computational-basis amplitudes are

$$\psi_c(z) = \langle z| |\psi_c\rangle = \langle z| H^{\otimes n} |\phi(c)\rangle. \tag{4}$$

The squared amplitudes $|\psi_c(z)|^2$ give the reference sampling distribution for condition c. Our requirements concern classical access to this state rather than the encoder's gate structure. Consider the bit masks $a \in \{0,1\}^n$ that index the Pauli-Z moments entering the objective (Sec. II C). For each condition c at which we train or evaluate the model, we require two efficient classical operations:

1. Sample $z \sim |\psi_c(z)|^2$.
2. Evaluate the amplitude ratio

$$R_c(z, a) = \frac{\psi_c(z \oplus a)}{\psi_c(z)} \tag{5}$$

for sampled z, where $\oplus$ denotes bitwise addition modulo two.

We call these two operations sampling-and-ratio access.

Computationally tractable reference states, in the sense of Van den Nest, admit efficient computational-basis sampling and amplitude evaluation [23]. These states provide a sufficient realization of sampling-and-ratio access, provided that their amplitudes can be evaluated with sufficient precision to compute the required ratios efficiently. Sampling-and-ratio access requires less than computational tractability, because amplitude ratios do not depend on the normalization of ψ_c. Havlíček argues that this access model is strictly weaker than sample-and-query access while retaining many of its simulation capabilities [33]. Nevertheless, our unbiasedness guarantees still require exact samples from $|\psi_c|^2$. In practice, all reference states used in this work are computationally tractable. The simplest example is a product-state encoder $|\phi(c)\rangle = \bigotimes_{i=1}^{n} |\phi_i(c)\rangle$. The corresponding reference amplitudes factorize as $\psi_c(z) = \prod_i \psi_{c,i}(z_i)$, where $\psi_{c,i}(b) = \langle b| H |\phi_i(c)\rangle$. Reference samples can therefore be generated by drawing each bit independently, while the amplitude ratio takes the form

$$R_c(z,a) = \prod_{i:a_i=1} \frac{\psi_{c,i}(1-z_i)}{\psi_{c,i}(z_i)}. \tag{6}$$

When the local amplitudes can be evaluated at constant arithmetic cost, both operations require at most $O(n)$ local operations. Beyond this product-state example, Sec. II E introduces correlated reference states that satisfy the same access requirements while enforcing exact output symmetries.

However, efficient sampling-and-ratio access alone does not guarantee that the model can solve a given task, as the encoder also constrains how the output probabilities depend on the condition c. For rotation encodings with angles linear in c, the encoding generators determine the accessible Fourier frequencies, while the learned parameters control their coefficients within the constraints imposed by the circuit architecture [34]. Multidimensional Fourier analyses further characterize how the encoding and circuit architecture constrain the model's expressive capacity [35]. Moreover, Caro *et al.* derived encoding-dependent generalization bounds for parameterized quantum circuits [36]. These results indicate that the encoder is a central modelling choice, alongside the circuit architecture.

B. Encoders equivalent to diagonal conditioning

Not every encoder adds conditioning beyond the diagonal layer. Let $\mathcal{G}$ denote the set of Pauli-Z generators of D_G, so that $D_G(\boldsymbol{\theta}) = \exp(i \sum_{g\in\mathcal{G}} \theta_g g)$.

Lemma 1 *Let two encoders ϕ and ϕ' have reference states related by $|\psi'_c\rangle = \Lambda_c |\psi_c\rangle$, where $\Lambda_c = \exp[i \sum_{g\in\mathcal{G}} \lambda_g(c)\, g]$. Then $U_{\boldsymbol{\theta}} |\phi'(c)\rangle = U_{\boldsymbol{\theta}+\boldsymbol{\lambda}(c)} |\phi(c)\rangle$ for every condition and parameter vector.*

Proof. Diagonal operators commute, so $D_G(\boldsymbol{\theta})\Lambda_c = D_G(\boldsymbol{\theta}+\boldsymbol{\lambda}(c))$. Hence $U_{\boldsymbol{\theta}} |\phi'(c)\rangle = H^{\otimes n} D_G(\boldsymbol{\theta}) \Lambda_c |\psi_c\rangle = U_{\boldsymbol{\theta}+\boldsymbol{\lambda}(c)} |\phi(c)\rangle$. □

If $\boldsymbol{\lambda}$ does not depend on c, the two encoders define the same model family up to a fixed shift of the parameters, and training cannot distinguish them. If $\boldsymbol{\lambda}$ depends on c, input conditioning reduces to condition-dependent diagonal parameters $\boldsymbol{\theta} + \boldsymbol{\lambda}(c)$. For example, the product encoder $\bigotimes_i R_X(t_i(c)) |0\rangle$ gives $|\psi_c\rangle = \bigotimes_i R_Z(t_i(c)) |+\rangle$, so it only shifts the one-body parameters.

The lemma also identifies what a diagonal operator cannot change. Every diagonal unitary preserves the reference magnitudes $|\psi_c(z)|$. Conversely, a reference state with uniform magnitudes $|\psi_c(z)| = 2^{-n/2}$ has the form $\Lambda_c |+\rangle^{\otimes n}$ for some diagonal Λ_c, possibly with generators outside $\mathcal{G}$. The conditional circuit then equals a standard IQP circuit with an additional condition-dependent diagonal layer. Therefore, an encoder conditions the model beyond diagonal phases only if its reference magnitudes are non-uniform. In that case, $U_{\boldsymbol{\theta}} |\phi(c)\rangle$ is not of the form $H^{\otimes n} D |+\rangle^{\otimes n}$ for any diagonal D, so no IQP circuit acting on $|0\rangle^{\otimes n}$ prepares it.

We call an encoder a phase encoder when its inputs have uniform computational-basis probabilities. Each such input takes the form $D_c |+\rangle^{\otimes n}$ for a diagonal D_c, so the reference state is itself an IQP state. A phase encoder alone gives the same uniform output distribution at every condition. Its conditional circuit applies two diagonal layers in different bases, and when its reference magnitudes are non-uniform, no choice of condition-dependent diagonal parameters reproduces its output states. This statement concerns states. It does not exclude that another model reproduces the same output distributions. Any condition dependence of the output therefore arises because the shared IQP unitary converts input phases into probabilities.

A reference state outside the IQP-state class therefore requires non-uniform input probabilities. This property is necessary but not sufficient for conditioning beyond diagonal parameters. Suppose that $|\phi(c)\rangle = U_{\boldsymbol{\lambda}(c)} |\phi'(c)\rangle$ for an encoder ϕ' and parameters $\boldsymbol{\lambda}(c)$ in the ansatz span. The reference state of ϕ is Λ_c applied to that of ϕ', so by the lemma the model equals the ϕ' model with parameters $\boldsymbol{\theta}+\boldsymbol{\lambda}(c)$. When ϕ' is a phase encoder, the input $|\phi(c)\rangle$ generally has non-uniform probabilities, yet the model is a phase-encoder model. The product encoder above is the simplest instance of this reduced form, with the fixed input $\phi' = |0\rangle^{\otimes n}$. The binomial and two-coordinate experiments in Sec. III use phase encoders. For the Curie–Weiss model, we construct an encoder whose input probabilities are non-uniform and which is not constructed in this reduced form, and we compare it with a phase encoder under the same protocol.

C. Conditional MMD objective

We train the shared parameter vector $\boldsymbol{\theta}$ by minimizing the squared maximum mean discrepancy (MMD) across the training conditions. For distributions p and q on $\{0,1\}^n$ and a positive-definite kernel k, the population squared MMD is [25]

$$\begin{aligned}\mathrm{MMD}^2(p,q) = \mathbb{E}_{x,x'\sim p}[k(x,x')] - 2\mathbb{E}_{x\sim p,\, y\sim q}[k(x,y)] \\ + \mathbb{E}_{y,y'\sim q}[k(y,y')],\end{aligned} \tag{7}$$

where we draw the two samples in each expectation independently.

Throughout this work, we use the Gaussian kernel

$$k_\sigma(x,y) = \exp\left[-\frac{d_H(x,y)}{2\sigma^2}\right]. \tag{8}$$

Here, $d_H(x,y)$ denotes the Hamming distance, which counts the positions at which x and y differ, and $\sigma > 0$ is the kernel bandwidth. For binary strings, $d_H(x,y) = \|x-y\|_2^2$, so the kernel corresponds to the standard Gaussian kernel restricted to the binary domain. For any finite $\sigma > 0$, the MMD vanishes if and only if $p = q$ [25].

Let $\mathcal{C}_{\text{train}}$ denote the set of training conditions and Σ the set of kernel bandwidths. The condition-balanced objective is

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{C}_{\text{train}}|}\sum_{c\in\mathcal{C}_{\text{train}}}\frac{1}{|\Sigma|}\sum_{\sigma\in\Sigma}\mathrm{MMD}^2_\sigma(p(\cdot\mid c), q_{\boldsymbol{\theta}}(\cdot\mid c)). \tag{9}$$

Here, MMD^2_σ denotes the MMD computed with the kernel k_σ. We weight all training conditions and bandwidths equally. At each optimizer update, we independently draw one condition uniformly from $\mathcal{C}_{\text{train}}$ and one bandwidth uniformly from Σ, giving a stochastic estimate of this average.

We express the objective through Pauli-Z moments, applying the Gaussian-kernel expansion into binary characters used in earlier QCBM and classical IQP training work [26, 27] separately at each condition. We sample a bit mask $a \in \{0,1\}^n$ by drawing its coordinates independently according to

$$a_i \sim \mathrm{Bernoulli}(p_\sigma), \qquad p_\sigma = \frac{1-e^{-1/(2\sigma^2)}}{2}. \tag{10}$$

The corresponding binary character is

$$\chi_a(x) = (-1)^{a\cdot x}, \tag{11}$$

which records the parity of the bits selected by a. For each bit where x and y agree, the expected character-product factor is 1. Where they differ, it is $1-2p_\sigma = e^{-1/(2\sigma^2)}$. Since the mask coordinates are independent, multiplying these factors recovers the Gaussian kernel:

$$k_\sigma(x,y) = \mathbb{E}_a[\chi_a(x)\chi_a(y)]. \tag{12}$$

For a condition c, define the target and model character moments

$$\begin{aligned}\mu_p(a\mid c) &= \mathbb{E}_{x\sim p(\cdot|c)}[\chi_a(x)], \\ \mu_q(a\mid c) &= \mathbb{E}_{x\sim q_{\boldsymbol{\theta}}(\cdot|c)}[\chi_a(x)].\end{aligned} \tag{13}$$

Substituting Eq. (12) into Eq. (7) gives

$$\mathrm{MMD}^2_\sigma(p(\cdot\mid c), q_{\boldsymbol{\theta}}(\cdot\mid c)) = \mathbb{E}_a\left[(\mu_p(a\mid c) - \mu_q(a\mid c))^2\right]. \tag{14}$$

Thus, the squared MMD averages the squared differences between target and model character moments over the mask distribution.

For a Born machine, the Pauli operator

$$Z_a = \bigotimes_{i=1}^n Z_i^{a_i} \tag{15}$$

has eigenvalue $\chi_a(x)$ on the computational-basis state $|x\rangle$. Its expectation value under the model is therefore

$$\mu_q(a\mid c) = \langle Z_a\rangle_{q_{\boldsymbol{\theta}}(\cdot|c)}. \tag{16}$$

We can thus express the conditional squared MMD entirely in terms of Pauli-Z moments, which we estimate classically in Sec. II D 1.

The bandwidth σ determines which correlation orders the objective emphasizes. The mask's Hamming weight $|a|$ counts the bits it selects, so it gives the correlation order of Z_a. Because we sample the mask coordinates independently,

$$\begin{aligned}|a| &\sim \mathrm{Binomial}(n, p_\sigma), \\ \mathbb{E}|a| &= np_\sigma, \\ \mathrm{Var}\,|a| &= np_\sigma(1-p_\sigma).\end{aligned} \tag{17}$$

Each bandwidth therefore induces a distribution over correlation orders. Large bandwidths favor lower-order correlations, whereas decreasing σ shifts weight toward higher-order correlations, with $\mathbb{E}|a| \to n/2$ as $\sigma \to 0$.

The bandwidth can also influence optimization behavior, as objectives dominated by global terms can exhibit barren plateaus [37–39]. Emphasizing lower-order terms can avoid this regime but reduces the sensitivity of the loss to higher-order differences between distributions [26], so the trained model may fail to capture them. Proposed mitigation strategies include tailored generator sets, restricted initializations and adversarial kernel adaptation [40–42]. Determining how these approaches extend to the conditional setting remains beyond the scope of this work.

D. Classical estimation of the objective and its gradients

1. *Conditional Pauli moments and phase derivatives*

Equation (14) expresses the objective through the conditional moments $\mu_q(a\mid c)$. We now use sampling-and-

ratio access to estimate these moments and their derivatives with respect to the parameters of the IQP unitary. Van den Nest established efficient sparse-observable estimation for computationally tractable states [23], while Havlíček developed estimation methods based on sampling and amplitude-ratio access [33]. Recio-Armengol *et al.* applied classical Pauli-moment estimation to MMD training of IQP Born machines with a fixed input [27]. Here, we apply these methods to prescribed condition-dependent reference states, enabling joint training of a single shared IQP unitary across conditions.

We write the diagonal layer as $D_{\boldsymbol{\theta}} = D_G(\boldsymbol{\theta})$ and index its $P = |V| + |E|$ generators by j, so that

$$D_{\boldsymbol{\theta}} |z\rangle = e^{i f_{\boldsymbol{\theta}}(z)} |z\rangle , \tag{18}$$

where

$$f_{\boldsymbol{\theta}}(z) = \sum_{j=1}^{P} \theta_j \chi_{g_j}(z), \qquad \chi_{g_j}(z) = (-1)^{g_j \cdot z}, \tag{19}$$

P denotes the number of generators, and $g_j \in \{0,1\}^n$ specifies the support of the jth Pauli-Z generator. The dot product $g_j \cdot z$ is evaluated modulo two. We define the phase difference associated with a bit mask a as

$$\Delta_a f_{\boldsymbol{\theta}}(z) = f_{\boldsymbol{\theta}}(z \oplus a) - f_{\boldsymbol{\theta}}(z). \tag{20}$$

The final Hadamard layer transforms Z_a into $X_a = \bigotimes_i X_i^{a_i}$. Consequently, the conditional Pauli moment can be expressed as

$$\begin{aligned} \mu_q(a \mid c) &= \langle \psi_c | \, D_{\boldsymbol{\theta}}^\dagger X_a D_{\boldsymbol{\theta}} \, |\psi_c\rangle \\ &= \sum_z \psi_c(z)^* \psi_c(z \oplus a) \\ &\quad \times e^{i \Delta_a f_{\boldsymbol{\theta}}(z)}. \end{aligned} \tag{21}$$

On the support of the reference sampling distribution, we can write

$$\psi_c(z)^* \psi_c(z \oplus a) = |\psi_c(z)|^2 R_c(z, a). \tag{22}$$

The sum therefore becomes an expectation over reference samples, with each amplitude ratio multiplied by the corresponding IQP phase factor. Since the Pauli moment is real, we take the real part of this product. Sampling never selects strings with $\psi_c(z) = 0$, so we evaluate amplitude ratios only on the support of $|\psi_c|^2$.

For a reference sample $z \sim |\psi_c|^2$, we define

$$\begin{aligned} W_{c,a}(z; \boldsymbol{\theta}) &= R_c(z, a) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)}, \\ Y_{c,a}(z; \boldsymbol{\theta}) &= \mathrm{Re}\, W_{c,a}(z; \boldsymbol{\theta}). \end{aligned} \tag{23}$$

The derivative of the phase difference with respect to θ_j is

$$\begin{aligned} d_{j,a}(z) &= \partial_{\theta_j} \Delta_a f_{\boldsymbol{\theta}}(z) \\ &= \chi_{g_j}(z \oplus a) - \chi_{g_j}(z) \in \{-2, 0, 2\}. \end{aligned} \tag{24}$$

For each reference sample, we define the phase-gradient estimator as

$$Y'_{j,c,a}(z; \boldsymbol{\theta}) = \mathrm{Re}[i \, d_{j,a}(z) W_{c,a}(z; \boldsymbol{\theta})] \, . \tag{25}$$

The relevant observables are

$$\begin{aligned} O_a &= D_{\boldsymbol{\theta}}^\dagger X_a D_{\boldsymbol{\theta}}, \\ \partial_{\theta_j} O_a &= i[O_a, Z_{g_j}]. \end{aligned} \tag{26}$$

Both operators are Hermitian and have at most one nonzero entry per row. The real parts of their local sparse-observable estimators are $Y_{c,a}$ and $Y'_{j,c,a}$, respectively. Since the reference state is independent of $\boldsymbol{\theta}$, these estimators satisfy, under exact sampling and arithmetic,

$$\begin{aligned} \mathbb{E}[Y_{c,a}] &= \mu_q(a \mid c), \\ \mathbb{E}[Y'_{j,c,a}] &= \partial_{\theta_j} \mu_q(a \mid c). \end{aligned} \tag{27}$$

The operator norm bounds $\|O_a\| = 1$ and $\|\partial_{\theta_j} O_a\| \le 2$ imply second moments bounded by one and four, respectively [33, Sec. 3.4]. In particular,

$$\mathbb{E}[|W_{c,a}|^2] = \sum_{z : \psi_c(z) \neq 0} |\psi_c(z \oplus a)|^2 \le 1. \tag{28}$$

Averaging S independent samples therefore yields mean-square errors of at most $1/S$ and $4/S$ for a given moment and phase derivative, respectively. By Chebyshev's inequality, $O(\epsilon^{-2})$ samples suffice to achieve additive error ϵ at fixed confidence. Each sample requires one amplitude ratio and the phase difference $\Delta_a f_{\boldsymbol{\theta}}(z)$, which involves only generators whose support overlaps a in an odd number of positions and costs at most $O(Pn)$ bit operations.

In practice, we draw S independent reference samples $z_s \sim |\psi_c|^2$ and estimate the moment as

$$\widehat{\mu}_q(a \mid c) = \frac{1}{S} \sum_{s=1}^{S} Y_{c,a}(z_s; \boldsymbol{\theta}). \tag{29}$$

Similarly, we estimate the phase derivative by averaging the corresponding $Y'_{j,c,a}$ values over the same samples. For the standard IQP input $|0\rangle^{\otimes n}$, all reference amplitudes are equal, giving $R_c(z, a) = 1$. The moment estimator then reduces to $Y = \cos[\Delta_a f_{\boldsymbol{\theta}}(z)]$, recovering the usual IQP cosine estimator [23, 43].

2. *Unbiased finite-sample MMD and gradients*

For a fixed condition c and mask a, we draw $B \ge 2$ independent target samples and compute

$$T_b = \chi_a(x_b), \qquad x_b \sim p(\cdot \mid c). \tag{30}$$

Independently of these target samples, we draw $S \ge 2$ independent reference samples and evaluate the reference-state estimator from Eq. (23),

$$Y_s = Y_{c,a}(z_s; \boldsymbol{\theta}), \qquad z_s \sim |\psi_c|^2. \tag{31}$$

We denote the corresponding sample means by

$$\overline{T} = \frac{1}{B}\sum_{b=1}^{B} T_b, \qquad \overline{Y} = \frac{1}{S}\sum_{s=1}^{S} Y_s. \tag{32}$$

We follow the independent-pair construction used for unbiased IQP MMD estimation [27, Appendix A.2.1], applying it to the conditional reference estimators and target moments. The direct estimate $(\overline{Y}-\overline{T})^2$ overestimates the squared moment difference on average because it includes the sampling variances of both sample means. We remove this bias by excluding products that use the same draw twice:

$$\begin{aligned}\widehat{\Delta_c^2}(a) = &\frac{1}{S(S-1)}\sum_{s\neq t} Y_s Y_t - 2\overline{Y}\,\overline{T} \\ &+ \frac{1}{B(B-1)}\sum_{b\neq b'} T_b T_{b'}.\end{aligned} \tag{33}$$

Independence within each pair gives

$$\mathbb{E}\left[\widehat{\Delta_c^2}(a)\right] = \left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)^2. \tag{34}$$

Thus, Eq. (33) provides an unbiased finite-sample estimator of the squared character-moment difference appearing in Eq. (14). An unbiased finite-sample estimate of MMD^2 may be negative. Once we have computed the sample values, we evaluate the estimator in time linear in the numbers of target and reference samples, using the pair-sum identities in Appendix A 2.

The same construction gives an unbiased estimator of the phase gradients. For a phase parameter θ_j, let

$$Y'_s = \partial_{\theta_j} Y_{c,a}(z_s;\boldsymbol{\theta}). \tag{35}$$

We evaluate Y'_s using Eq. (25). Differentiating Eq. (33) gives

$$\partial_{\theta_j}\widehat{\Delta_c^2}(a) = \frac{2}{S(S-1)}\sum_{s\neq t} Y_s Y'_t - 2\overline{T}\,\overline{Y'}, \tag{36}$$

where

$$\overline{Y'} = \frac{1}{S}\sum_{s} Y'_s. \tag{37}$$

Taking the expectation yields

$$\mathbb{E}\left[\partial_{\theta_j}\widehat{\Delta_c^2}(a)\right] = 2\left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)\partial_{\theta_j}\mu_q(a \mid c), \tag{38}$$

which is the gradient of the corresponding squared moment difference.

Since $|T_b| = 1$, independence of distinct draws and the triangle inequality in L^2 give

$$\begin{aligned}\mathbb{E}\left[\left(\widehat{\Delta_c^2}(a)\right)^2\right] &\leq 16, \\ \mathbb{E}\left[\left(\partial_{\theta_j}\widehat{\Delta_c^2}(a)\right)^2\right] &\leq 64.\end{aligned}$$

Averaging over conditions, bandwidths and sampled masks preserves these uniform bounds. Therefore, averaging N independent repetitions of the complete estimator gives mean-square error at most $16/N$ for the objective and $64/N$ for a specified phase-gradient component. By Chebyshev's inequality, $O(\epsilon^{-2})$ repetitions suffice for additive error ϵ at fixed confidence.

3. *Mask sampling*

Equation (14) expresses the squared MMD as an expectation over the mask distribution in Eq. (10). Since there are 2^n possible masks, we estimate this expectation using a finite sample of masks.

The identity mask $a = 0$ satisfies

$$\mu_p(0 \mid c) = \mu_q(0 \mid c) = 1 \tag{39}$$

for every condition and therefore contributes zero to the squared MMD. We consequently sample only nonidentity masks from the conditional distribution

$$P_\sigma(a \mid a \neq 0). \tag{40}$$

Under the original mask distribution, nonidentity masks have total probability

$$Q_\sigma^+ = 1 - (1-p_\sigma)^n. \tag{41}$$

We multiply the conditional expectation by Q_σ^+ to preserve the original Gaussian kernel objective.

We draw M nonidentity masks $a_1, \ldots, a_M$ independently from $P_\sigma(a \mid a \neq 0)$ and estimate the squared MMD at condition c as

$$\widehat{\mathrm{MMD}}_\sigma^2(c) = Q_\sigma^+ \frac{1}{M}\sum_{m=1}^{M} \widehat{\Delta_c^2}(a_m). \tag{42}$$

Here, $\widehat{\Delta_c^2}$ is the unbiased finite-sample estimator in Eq. (33). For each sampled condition and bandwidth, we use the same target and reference batches for all masks, drawing these batches independently of the masks. Sharing batches correlates the estimates across masks, but linearity of expectation preserves unbiasedness. Averaging over the mask, target and reference draws therefore recovers the full kernel squared MMD. Appendix A 2 describes an equivalent evaluation that groups repeated masks.

E. Input-state-induced symmetries

The freedom to choose correlated input states also allows the encoder to impose exact output symmetries. For a bit mask $h \in \{0,1\}^n$, the operator $X_h = \bigotimes_{i=1}^n X_i^{h_i}$ commutes with the IQP unitary. If the encoded input is an eigenstate of X_h, applying the unitary preserves its eigenvalue. The output amplitudes at x and $x \oplus h$

then differ at most by a sign, so their probabilities are equal. This principle underlies the input-state construction of Recio-Armengol *et al.* for a fixed input, who also outlined its extension to other commuting symmetries [27, Sec. 6]. Such symmetry constraints provide inductive biases in variational quantum models [44]. We now construct condition-dependent reference states that enforce several prescribed bit-flip symmetries while retaining sampling-and-ratio access.

These reference states have computational-basis configurations that satisfy prescribed parity constraints. Let $h_1, \ldots, h_r \in \mathbb{F}_2^n$ be linearly independent masks specifying the desired bit-flip symmetries. Choose a matrix $C \in \mathbb{F}_2^{n\times k}$, with $k = n - r$, whose columns form a basis of the subspace orthogonal to all these masks. For any string of latent bits $y \in \mathbb{F}_2^k$, the reference configuration Cy therefore satisfies $h_j^\top Cy = 0$ for every j. All matrix operations below are over $\mathbb{F}_2$.

For each condition c and latent bit $\ell = 1, \ldots, k$, choose efficiently evaluable complex amplitudes $f_\ell(0; c)$ and $f_\ell(1; c)$ satisfying

$$|f_\ell(0; c)|^2 + |f_\ell(1; c)|^2 = 1. \tag{43}$$

We define the reference state and encoded input as

$$\begin{aligned} |\psi_c\rangle &= \sum_{y\in\mathbb{F}_2^k} \left[\prod_{\ell=1}^{k} f_\ell(y_\ell; c)\right] |Cy\rangle\,, \\ |\phi(c)\rangle &= H^{\otimes n}\,|\psi_c\rangle\,. \end{aligned} \tag{44}$$

We refer to this construction as a binary-linear encoder. The matrix C fixes the parity constraints, while the factor amplitudes encode the condition. Since C is injective, distinct latent strings give distinct reference configurations, and the factor normalization ensures that the reference state is normalized.

To sample the reference distribution, we draw each latent bit independently with probability $|f_\ell(1; c)|^2$ of obtaining one and return Cy. If a mask a belongs to $\operatorname{im} C$, there is a unique latent displacement $d \in \mathbb{F}_2^k$ satisfying $a = Cd$. Flipping the reference bits selected by a then maps Cy to $C(y \oplus d)$, giving

$$R_c(Cy, a) = \prod_{\ell=1}^{k} \frac{f_\ell(y_\ell \oplus d_\ell; c)}{f_\ell(y_\ell; c)}. \tag{45}$$

If $a \notin \operatorname{im} C$, the flipped configuration lies outside the reference support and the ratio is zero. We evaluate denominators only at sampled configurations with nonzero amplitude.

The space $\ker C^\top$ is the span of the prescribed symmetry masks. For every h in this space, the parity constraints imply $Z_h\,|\psi_c\rangle = |\psi_c\rangle$, and the Hadamard layer gives $X_h\,|\phi(c)\rangle = |\phi(c)\rangle$. Since $U_{\boldsymbol{\theta}}$ commutes with X_h, the output retains this eigenvalue. Consequently,

$$q_{\boldsymbol{\theta}}(x \mid c) = q_{\boldsymbol{\theta}}(x \oplus h \mid c) \qquad \text{for all } h \in \ker C^\top. \tag{46}$$

The construction therefore guarantees a group of 2^r bit-flip symmetry transformations for every condition and parameter vector. Particular factor amplitudes or parameter values may produce additional symmetries. Reference sampling and amplitude-ratio evaluation have polynomial cost, provided that the factor amplitudes admit efficient evaluation with sufficient precision. Appendix C establishes normalization, access costs and state preparation.

The simplest instance enforces global bit-flip symmetry, $h = 1^n$. Taking $Cy = (y, \bigoplus_\ell y_\ell)$, with $k = n - 1$, restricts the reference state to even-parity configurations, so that $q_{\boldsymbol{\theta}}(x \mid c) = q_{\boldsymbol{\theta}}(\bar{x} \mid c)$, where $\bar{x}$ flips all bits. We call this construction the parity encoder. Section III C uses it to encode inverse temperature and extends it beyond phase encoding. Appendix D gives a sixteen-qubit example that enforces six independent switching symmetries of signed graphs at once.

III. NUMERICAL EXPERIMENTS

We evaluate conditional IQP models at sixteen qubits, where exact output probabilities are available, and assess computational feasibility by training models from 64 to 2048 qubits. We train all models by minimizing the sampled squared MMD objective. We use hyperparameter optimization for the two-coordinate, Curie–Weiss and scaling benchmarks. Appendices A and B describe the training settings, hyperparameter searches and evaluation protocols. Code and data are available in the accompanying repository [45].

Each experiment uses three disjoint sets of conditions. Training conditions define the objective. Development conditions select checkpoints and hyperparameters. Because they are excluded from training, this selection optimizes performance at held-out conditions, as is standard in machine learning. Test conditions never enter training or selection and provide the reported results. At sixteen qubits, we select on exact full-distribution TVD at the development conditions. Because this selection uses exact target probabilities, which are unavailable in practice, the reported accuracies indicate what the model can attain rather than what purely sample-based selection would achieve. At larger sizes, we select on sampled MMD at the training conditions. The target families, condition ranges and encoder settings were developed through earlier exploratory runs, so the experiments assess conditional generation within these families.

For the sixteen-qubit experiments, our primary accuracy metric is the exact full-distribution total variation distance,

$$\mathrm{TVD}(p, q) = \frac{1}{2} \sum_{x\in\{0,1\}^n} |p(x) - q(x)|. \tag{47}$$

We also compare the Hamming-weight profiles using

$$\mathrm{TVD}_K(p,q) = \frac{1}{2}\sum_{k=0}^{n} |\rho_p(k) - \rho_q(k)|, \tag{48}$$

where $\rho_p(k)$ and $\rho_q(k)$ are the probabilities of observing Hamming weight k. This gives a coarser measure of agreement, with $\mathrm{TVD}_K(p,q) \le \mathrm{TVD}(p,q)$. Unless stated otherwise, we compute each metric within a fit, average it over the relevant conditions, and report the mean and sample standard deviation across independent fits.

A. Conditional generation of binomial distributions

As an initial validation of the conditional generation framework, we consider a family of binomial distributions over Hamming weight. In this benchmark, the conditional IQP model must learn the dependence of the target distribution on an input condition $c \in [0,1]$, which smoothly controls its mean Hamming weight.

For $n = 16$ bits, we denote the Hamming weight by $K = |x|$ and define the target profile as

$$\rho(k \mid c) = \binom{16}{k} r(c)^k [1 - r(c)]^{16-k}, \tag{49}$$

where $k \in \{0, \ldots, 16\}$ and

$$r(c) = \frac{1}{1 + \exp[-\frac{3}{4}(2c-1)]}. \tag{50}$$

The target mean Hamming weight is $16r(c)$, which changes smoothly with c. Its variance is $16r(c)[1-r(c)]$, so varying the condition also changes the width of the profile. We assign equal probability to bitstrings with the same Hamming weight,

$$p(x \mid c) = \frac{\rho(|x| \mid c)}{\binom{16}{|x|}}. \tag{51}$$

Equivalently, the target treats the bits as independent, each equal to one with probability $r(c)$, so this benchmark tests conditional control of a product distribution.

We use a product-state encoder that represents the condition through input phases,

$$|\phi(c)\rangle = \bigotimes_{i=1}^{16} R_Z(c - 1/2) R_Y(\pi/2) \, |0\rangle_i \, . \tag{52}$$

This choice gives uniform computational-basis probabilities for every condition. The learned IQP unitary must therefore convert the encoded phases into the condition-dependent target probabilities. At $c = 0.5$, the encoder prepares $|+\rangle^{\otimes 16}$, which the IQP unitary preserves up to a global phase. The resulting uniform bitstring distribution exactly matches the target at this condition for every parameter vector, so agreement at the midpoint holds by construction.

We parameterize the learned IQP unitary using a 4×4 lattice with sixteen one-body and twenty-four nearest-neighbor two-body generators, assigning an independent parameter to each generator. We train at $c \in \{0, 0.25, 0.5, 0.75, 1\}$, select checkpoints at four intermediate development conditions, and test at 80 conditions distributed throughout $[0,1]$. We perform five independent fits. Appendix B 1 describes the training settings and checkpoint selection.

Figure 2 shows that the model reproduces the change in the target profiles and follows the target mean between training conditions. Averaged over the 80 test conditions, the full-distribution TVD is 0.0649 ± 0.0053, compared with 0.2848 for the encoder alone. The profile TVD is 0.0200 ± 0.0032. The excess of the full-distribution error over the profile error arises from how the model distributes probability among bitstrings of equal Hamming weight. Table I summarizes these errors.

TABLE I. Exact full-distribution and profile TVDs for conditional generation of binomial distributions. Entries report the mean error over eighty test conditions, given as the mean and sample standard deviation across five independent fits. Lower is better.

Model	Full TVD	Profile TVD
Conditional IQP	0.0649 ± 0.0053	0.0200 ± 0.0032
Encoder only	0.2848	0.2848

These results show that the model learns the relation between a single input condition and the target distribution within the tested interval. We next consider a two-coordinate target family to test control of both mean and width, including combinations of conditions absent from training.

B. Two-coordinate conditioning and generalization

We now extend the binomial benchmark by introducing a second conditioning variable that changes the width of the target distribution beyond the constraint imposed by the binomial family. The aim is to test whether the conditional IQP model can learn the joint dependence on both variables and generate distributions at combinations absent from training. We train on conditions that vary one coordinate at a time, so that we can test generalization to joint conditions beyond the convex hull of the training set.

For conditions $u, v \in [-1,1]$, we define the target Hamming-weight profile as

$$\rho_{u,v}(k) \propto \binom{16}{k} \exp\left[-\frac{(k-8-3u)^2}{8(1+0.2v)^2} + \frac{(k-8)^2}{8}\right], \tag{53}$$

with normalization over $k \in \{0, \ldots, 16\}$. As in the previous benchmark, we assign equal probability to bitstrings

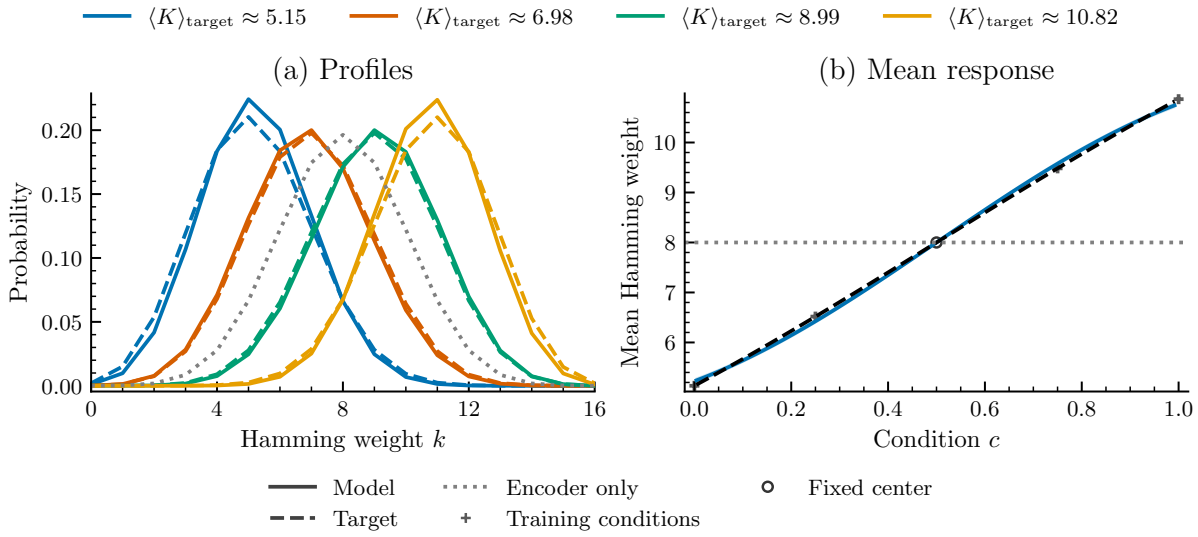


FIG. 2. Conditional generation of binomial distributions at sixteen qubits. (a) Solid colored curves show exact model Hamming-weight profiles averaged across five independent fits at four test conditions, $c \approx \{0.004, 0.329, 0.667, 0.992\}$. Dashed curves of the same color show the corresponding targets. The upper labels give the target mean Hamming weight for each color. (b) The solid blue curve and shading show the mean and one sample standard deviation of the model's mean Hamming weight across five fits at 80 test conditions. The dashed black curve shows the target mean. Dotted gray curves show the encoder-only binomial profile and constant mean; gray crosses mark training conditions. The open circle at $c = 0.5$ marks agreement that holds by construction.

with the same Hamming weight,

$$p(x \mid u, v) = \frac{\rho_{u,v}(|x|)}{\binom{16}{|x|}}. \tag{54}$$

The coordinate u shifts the profile, while v changes its width. Both coordinates affect the actual mean and standard deviation, which are not given directly by $8+3u$ and $2(1+0.2v)$. We therefore evaluate the model against the mean and standard deviation computed from the target distribution. Setting $v = 0$ and $u = 2c - 1$ recovers the binomial family from the previous benchmark. At $(u, v) = (0, 0)$, the target bitstring distribution is uniform, with a binomial Hamming-weight profile of mean 8 and standard deviation 2.

We extend the previous phase encoder by encoding u through single-qubit Z rotations and v through pairwise ZZ rotations. Defining $J_z = \frac{1}{2}\sum_{j=1}^{16} Z_j$, we prepare

$$|\phi(u, v)\rangle = e^{-i(u/2)J_z} e^{-i(v/40)J_z^2} |+\rangle^{\otimes 16}. \tag{55}$$

At $v = 0$, this recovers the product-state encoder of the previous benchmark under $u = 2c - 1$. The additional pairwise rotations generally produce correlated input states. Both operations change only the input phases, leaving the computational-basis probabilities uniform for every condition. The learned IQP unitary must therefore convert these phases into the varying target probabilities. At $(u, v) = (0, 0)$, the input is $|+\rangle^{\otimes 16}$, so, as in the binomial benchmark, the model matches the uniform target by construction.

The encoded amplitudes depend only on Hamming weight, and the reference state $H^{\otimes 16} |\phi(u, v)\rangle$ retains this property. This structure allows efficient classical sampling and amplitude-ratio evaluation despite the input correlations. Appendix B 2 describes these operations and the circuit preparation.

We retain the same IQP unitary architecture as in the binomial benchmark, with forty independent parameters on a 4×4 lattice. This unitary need not preserve the permutation symmetry of the target distribution, so we evaluate accuracy using full-distribution TVD as well as Hamming-weight profile TVD.

We train the model using conditions on the two coordinate axes, which we refer to as axis training. These conditions have the form $(u, 0)$ or $(0, v)$, so only one coordinate varies at a time. We use nine distinct training conditions, including the origin. Their convex hull is the diamond-shaped region $|u| + |v| \leq 1$.

Development conditions lie on the same axes. Thus, neither training nor model selection uses mixed conditions, defined here as pairs with both $u \neq 0$ and $v \neq 0$. We test the selected models at sixteen mixed pairs. Each individual coordinate value appears in the training set, but their combinations do not. Four test pairs lie inside the training convex hull and twelve lie outside it. We perform five independent fits. Appendix B 2 specifies the search and evaluation protocol and reports a complementary grid-training experiment.

After axis training, the model approximates the target distributions at joint conditions excluded from both

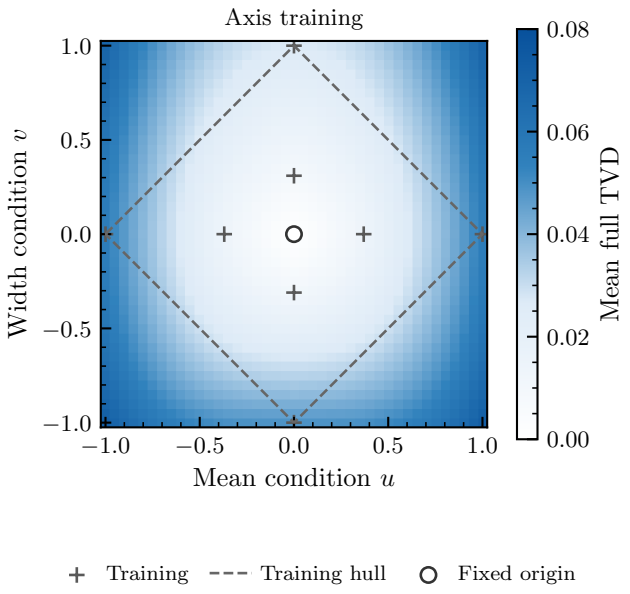


FIG. 3. Joint conditional generation after axis training. Colors show the mean full-distribution TVD across five independent fits on a 41×41 diagnostic grid. Gray crosses mark training conditions, and the dashed diamond marks their convex hull. The open circle at $(u, v) = (0, 0)$ marks the uniform target and model distribution fixed by construction. The diagnostic grid includes training and development conditions; quantitative results use sixteen separate mixed test conditions.

training and model selection. The sixteen mixed test pairs give a full-distribution TVD of 0.0468 ± 0.0143, compared with 0.3867 for the encoder alone (Table IV). The four pairs inside the training convex hull give 0.0185 ± 0.0060, while the twelve outside give 0.0563 ± 0.0171. The model therefore approximates joint target distributions beyond the training convex hull. The higher error outside the hull does not by itself isolate an extrapolation penalty, because the target conditions also differ between the two subsets.

Figure 4 shows how the models respond to each coordinate. The model mean closely follows the target across the tested widths. The model also captures the increase in width with v, although it overestimates the width of the narrowest target distributions. Averaged across fits, the model reproduces between 0.97 and 1.04 of the target change in mean along u, and between 0.83 and 0.92 of the target change in width along v (Appendix B 2). These results support joint conditional control of the mean and width, including at combinations absent from training.

C. Curie–Weiss spin model

We next test whether conditional IQP models can learn a temperature-dependent change in distribution shape. We consider the zero-field Curie–Weiss spin model [46], whose magnetization profile changes from a single central peak to two symmetric peaks as temperature decreases.

For $n = 16$ spins, we represent each configuration by a bitstring $x \in \{0,1\}^n$ and assign $s_i = 2x_i - 1$. The magnetization per spin is therefore $m(x) = n^{-1}\sum_i s_i = (2|x| - n)/n$. With unit interaction strength and inverse temperature β, the target distribution is

$$p(x \mid \beta) = \frac{1}{\mathcal{Z}_\beta} \exp\left[\frac{\beta}{2n}(2|x| - n)^2\right], \tag{56}$$

where $\mathcal{Z}_\beta$ normalizes the distribution. The target assigns equal probability to bitstrings with the same Hamming weight and is invariant under global spin flip.

We encode inverse temperature with a squeezed parity encoder, whose input probabilities are non-uniform. Its design follows the symmetries of the target. The target depends only on Hamming weight and is invariant under global spin flip, so the encoder uses collective operations that preserve permutation symmetry among the leaves and commute with $X^{\otimes16}$. A ZZ layer coupling one root qubit to every leaf provides the flip-symmetric correlations of the parity encoder used below. A phase encoder cannot change the input probabilities of Hamming weights, which remain binomial, so we add collective squeezing of the leaves, which redistributes probability among leaf Hamming weights at a cost of one parameter. We call qubit 16 the root and qubits 1 to 15 the leaves, and we write $J_\alpha = \frac{1}{2}\sum_{i=1}^{15}\sigma_i^\alpha$ for the collective spin of the leaves. The input state is

$$|\phi_S(\beta)\rangle = \exp\left[-\frac{it_\beta}{2}\sum_{i=1}^{15} Z_i Z_{16}\right]\left[S(r_\beta)\,|+\rangle^{\otimes15}\right] \otimes |+\rangle\,, \tag{57}$$

where

$$S(r) = \exp\left[\frac{ir}{15}\,(J_yJ_z + J_zJ_y)\right] \tag{58}$$

is a two-axis squeezing operation on the leaves [47], since $J_yJ_z + J_zJ_y = J_u^2 - J_v^2$ with $J_{u,v} = (J_y \pm J_z)/\sqrt{2}$. The angles follow the fixed schedule

$$\sin t_\beta = \tanh(a\beta^2), \qquad r_\beta = -\kappa\beta\cos t_\beta. \tag{59}$$

The ZZ angle t_β grows with β and saturates at $\pi/2$, where the ZZ layer equals a controlled-Z gate between the root and every leaf up to single-qubit phases, and the factor $\cos t_\beta$ switches the squeezing off in this limit. We fixed $\kappa = 0.4$ and the search range of a in a preliminary comparison of encoder variants on development temperatures, and we select a by hyperparameter optimization, so the encoder has no trainable parameters. The ZZ rotations correlate the root with every leaf. The squeezing redistributes probability among leaf Hamming weights, so the input probabilities are non-uniform for every $\beta > 0$ and the reference state is not an IQP state. Their departure from uniform, measured by TVD, rises from 0.076 at $\beta = 0.4$ to a maximum of 0.172 near $\beta = 1.26$ and falls

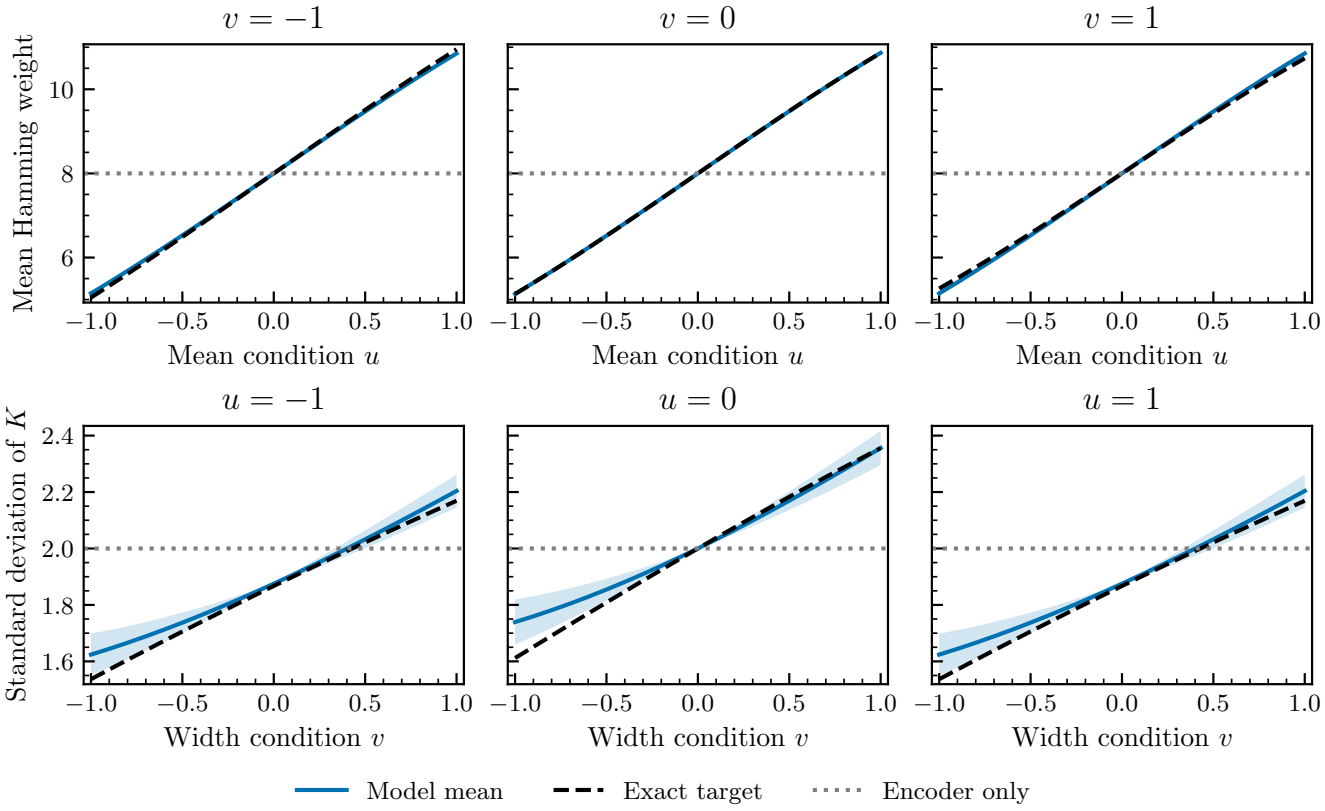


FIG. 4. Mean and width responses after axis training. Top row: mean Hamming weight as a function of u at $v = -1, 0, 1$. Bottom row: standard deviation of the Hamming weight K as a function of v at $u = -1, 0, 1$. Solid blue curves and shading show the mean and one sample standard deviation across five independent fits. Dashed black curves show exact target values, and dotted gray lines show the encoder-only values. Each slice contains 21 conditions, including training and development conditions.

to 0.104 at $\beta = 2$. The squeezing acts before the ZZ rotations and is diagonal in neither the computational nor the Hadamard basis, so the encoder is not constructed in the reduced form of Sec. II B. Appendix B 3 a tests numerically how close it comes to that form. At $\beta = 0$, the input is $|+\rangle^{\otimes 16}$, so the model matches the uniform target by construction.

Setting $r_\beta = 0$ and $t_\beta = \lambda\beta$ gives the parity encoder of Sec. II E,

$$|\phi_{\mathrm{P}}(\beta;\lambda)\rangle = \exp\left[-\frac{i\lambda\beta}{2}\sum_{i=1}^{15} Z_i Z_{16}\right]|+\rangle^{\otimes 16}, \qquad (60)$$

where λ is an encoder scale. This phase encoder leaves the input probabilities uniform, so its reference state is an IQP state. We use it as the main comparison for the squeezed parity encoder.

Both encoders enforce the target's global spin-flip symmetry. Every layer in Eq. (57) commutes with $X^{\otimes 16}$, because the global flip reverses both J_y and J_z and therefore preserves their symmetrized product. The input is thus an eigenstate of $X^{\otimes 16}$, and the argument of Sec. II E gives $q_{\boldsymbol{\theta}}(x \mid \beta) = q_{\boldsymbol{\theta}}(\bar{x} \mid \beta)$ for every parameter vector. This enforces zero signed mean magnetization and a symmetric magnetization profile. It does not impose permutation symmetry or agreement with the target probabilities. Every layer is also invariant under permutations of the leaves, so the reference amplitudes depend only on the root bit and the leaf Hamming weight. A table of 2×16 amplitudes per temperature, computed in polynomial time, therefore provides exact sampling and amplitude ratios. Neither encoder requires ancillary qubits. Appendix B 3 gives the amplitude table, the sampling procedure and the preparation circuits.

We parameterize each IQP unitary with all sixteen one-body and 120 two-body generators, giving 136 independent parameters per circuit. We train the conditional IQP model at $\beta \in \{0.1, 0.7, 1.3, 1.95\}$ and test it at $\beta \in \{0.4, 1, 2\}$ and on a separate dense grid.

We compare conditional IQP, which uses the squeezed parity encoder, with two kinds of models. First, conditional models with the parity input or a product input use the same IQP ansatz, training temperatures and budgets, so they isolate the effect of the encoder. The product encoder, defined in Appendix B 3 d, is also a phase encoder but does not enforce global spin-flip symmetry.

Second, three families of independently trained circuits each contain three circuits, one trained directly at each test temperature. The temperature-specific IQP models use the squeezed parity encoder, the parity-input models use the parity encoder, and the standard-input models use $|0\rangle^{\otimes 16}$. Within each family, the circuits have separate learned parameters but use one jointly selected hyperparameter configuration. Because temperature-specific models cannot generate at unseen conditions, each family trains one circuit per test temperature, which gives it three times the learned parameters and optimizer updates of the shared model. This comparison tests conditional generation at unseen temperatures against models trained specifically for those targets.

We optimize hyperparameters separately for each model family. For checkpoint and hyperparameter selection, we use exact probabilities to compute a full-distribution TVD score. Shared models use a development grid that excludes the test temperatures, whereas temperature-specific models are selected at their training temperatures, which coincide with the test temperatures. The development grid is dense and includes temperatures within 0.025 of each test temperature, but the test temperatures themselves enter neither training nor selection of the shared model. Among candidates within a fixed tolerance of the best score, we select the candidate with the lowest combined error in mean absolute magnetization and energy fluctuations. Appendix B 3 c defines these scores and specifies the optimization budgets and selection protocol. We perform ten independent final fits for each family.

Figure 5 shows that conditional IQP captures the change from a central magnetization peak in the hot regime ($\beta = 0.4$) to two peaks at large absolute magnetization in the cold regime ($\beta = 2$). Near the crossover at $\beta = 1$, it follows the broad target profile, whereas the parity-input model places too much probability at large $|m|$ and too little near $m = 0$. The test temperatures $\beta = 0.4$ and 1 require interpolation, while $\beta = 2$ requires slight extrapolation beyond the largest training inverse temperature, 1.95.

On the separate 101-point test grid in $[0, 2]$, we compute each fit's inverse-temperature-averaged full-distribution TVD and maximum TVD. Across ten fits, these are 0.0474 ± 0.0028 and 0.0677 ± 0.0070 for conditional IQP, compared with 0.1033 ± 0.0036 and 0.1406 ± 0.0037 for the parity-input model. The mean error of conditional IQP grows slowly with β and is largest at the extrapolated endpoint $\beta = 2$, whereas the parity-input model has its largest errors near the crossover.

Figure 6 isolates the effect of the encoder. Relative to the parity input, the squeezed parity encoder reduces the full-distribution TVD score E_{TVD} from 0.1281 ± 0.0027 to 0.0609 ± 0.0055 and lowers the mean TVD at every nonzero point of the test grid. Because both models draw their training data and stochastic estimates from the same random streams, we can also compare them fit by fit, and the squeezed parity encoder has the lower score in all ten paired fits. The product input, which enforces no spin-flip symmetry, gives the largest errors. Table VIII in Appendix B 3 d lists these scores. The squeezed and parity encoders enforce the same symmetry, so symmetry alone does not explain the difference between them. The squeezing itself drives the gain. Removing it while keeping the angle schedule and the selected settings raises E_{TVD} to 0.1692 ± 0.0073, above the parity input, in all ten paired fits (Appendix B 3 d).

Table II compares the models at the three test temperatures. Conditional IQP has full-distribution TVDs of 0.0366 ± 0.0016, 0.0555 ± 0.0043 and 0.0671 ± 0.0075 at $\beta = 0.4$, 1 and 2. Temperature-specific IQP uses the same encoder but trains a separate circuit at each temperature, and it has full-distribution TVDs of 0.0378 ± 0.0069, 0.0613 ± 0.0130 and 0.0494 ± 0.0054. The shared model therefore matches these circuits within one standard deviation at $\beta = 0.4$ and 1, but has higher TVD at $\beta = 2$. With the parity input, the temperature-specific circuits reach 0.0692 ± 0.0059, 0.1477 ± 0.0081 and 0.1525 ± 0.0021, so the choice of encoder affects accuracy more than the choice between one shared model and separate circuits. The standard-input baseline has the largest mean full-distribution TVD at all three temperatures.

We also examine the temperature dependence of mean absolute magnetization and energy fluctuations. We use $\langle |m| \rangle$ because both encoders enforce zero signed mean magnetization. For the energy $H = -nm^2/2$, we compute the per-spin energy-fluctuation statistic

$$c_{\mathrm{fluc}}(\beta) = \frac{\beta^2}{n}\big(\langle H^2 \rangle - \langle H \rangle^2\big). \tag{61}$$

For the Gibbs target, this expression gives the equilibrium heat capacity per spin. For learned distributions, we use it as an energy-fluctuation diagnostic. Appendix B 3 f details the calculations.

Figure 7 shows that conditional IQP follows the increase in mean absolute magnetization and the peaked energy-fluctuation response across the test grid. Its averaged fluctuation peak is 0.816 at $\beta = 1.54$, close to the target peak of 0.823 at $\beta = 1.50$. The parity-input model underestimates this peak, at 0.771, and places it at $\beta = 1.60$. Relative to the parity input, the squeezed parity encoder reduces the physical-observable score E_{phys} on the test grid from 0.0865 ± 0.0088 to 0.0181 ± 0.0030. The largest gain occurs in the cold regime, where the normalized fluctuation error at $\beta = 2$ falls from 0.171 ± 0.027 to 0.023 ± 0.014. With the same encoder, temperature-specific IQP has smaller magnetization errors at all three test temperatures. Its fluctuation error is smaller at $\beta = 2$, 0.0057 ± 0.0084 against 0.0234 ± 0.0138 for the shared model, but larger at $\beta = 1$, 0.0196 ± 0.0080 against 0.0087 ± 0.0018. The standard-input baseline varies substantially across fits at $\beta = 2$. Table X in Appendix B 3 f lists the observable errors.

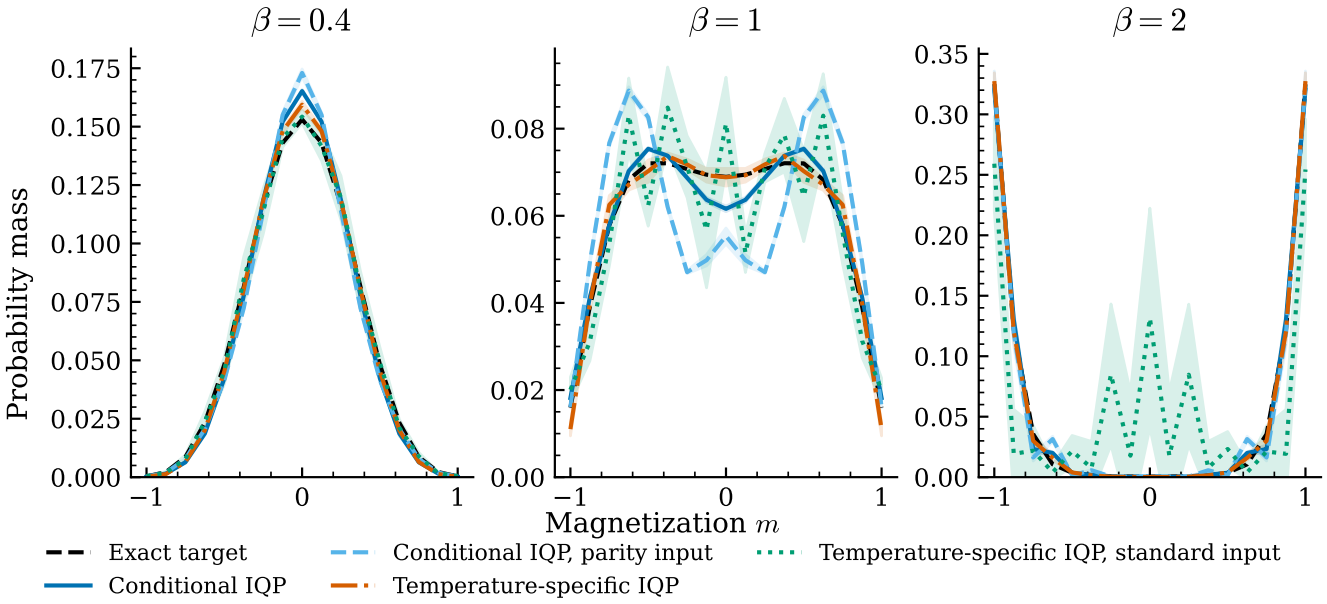


FIG. 5. Curie–Weiss magnetization profiles at sixteen qubits. Magnetization per spin is $m = (2k - 16)/16$, where k is Hamming weight. Colored curves show exact probability masses averaged across ten independent fits. Shading shows one sample standard deviation above and below the mean. Dashed black curves show the targets. Conditional IQP and temperature-specific IQP use the squeezed parity encoder, the parity-input model uses the parity encoder, and the standard-input baseline uses $|0\rangle^{\otimes 16}$. The displayed test temperatures are excluded from conditional training and model selection, while the temperature-specific models are trained directly at them. Panels use separate vertical scales. Table II reports full-distribution errors.

TABLE II. Exact full-distribution TVD at sixteen qubits. Values are means ± sample standard deviations over ten independent fits, computed within each fit before aggregation. Lower is better. Conditional and temperature-specific IQP use the squeezed parity encoder; the parity-input models use the parity encoder, and the standard-input model uses $|0\rangle^{\otimes 16}$. Conditional models exclude these test temperatures from training and selection, whereas temperature-specific models are trained directly at each temperature.

Model	$\beta = 0.4$	$\beta = 1$	$\beta = 2$
Conditional IQP	0.0366 ± 0.0016	0.0555 ± 0.0043	0.0671 ± 0.0075
Conditional IQP, parity input	0.1185 ± 0.0107	0.1396 ± 0.0048	0.0832 ± 0.0103
Temperature-specific IQP	0.0378 ± 0.0069	0.0613 ± 0.0130	0.0494 ± 0.0054
Temperature-specific IQP, parity input	0.0692 ± 0.0059	0.1477 ± 0.0081	0.1525 ± 0.0021
Temperature-specific IQP, standard input	0.3787 ± 0.0034	0.3400 ± 0.0074	0.4101 ± 0.1629

D. Scaling

We next assess the computational feasibility of classical conditional IQP training at larger system sizes. We train Curie–Weiss models with $n \in \{64, 128, 256, 512, 1024, 2048\}$ qubits using the parity encoder, without ancillas. This encoder enforces global spin-flip symmetry at every size, and its reference amplitudes factorize over latent bits. The amplitude table of the squeezed parity encoder also has polynomial cost, but we have not assessed its numerical stability at these sizes. We assign independent trainable phases to each qubit and each horizontal or vertical nearest-neighbor pair on an open rectangular lattice. An $L_x \times L_y$ lattice therefore has $3n - L_x - L_y$ trainable parameters.

We optimize hyperparameters at 64 and 256 qubits and use the selected configuration at all six sizes. We train at inverse temperatures $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and test at $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$. At each size, we perform three independent fits with three initializations each. Validation at the training conditions selects a checkpoint and its associated initialization within each fit, and we report the full training trajectory of that initialization. Appendix B 4 specifies the optimization and evaluation protocols.

We use two kernel bandwidths, chosen to give expected mask weights of two and four under the mask distribution in Eq. (10). This choice emphasizes low-order Pauli-Z moments at every system size, so the corresponding bandwidths increase with n. We assess performance using classical estimates of the squared MMD, averaged over the test temperatures and both bandwidths.

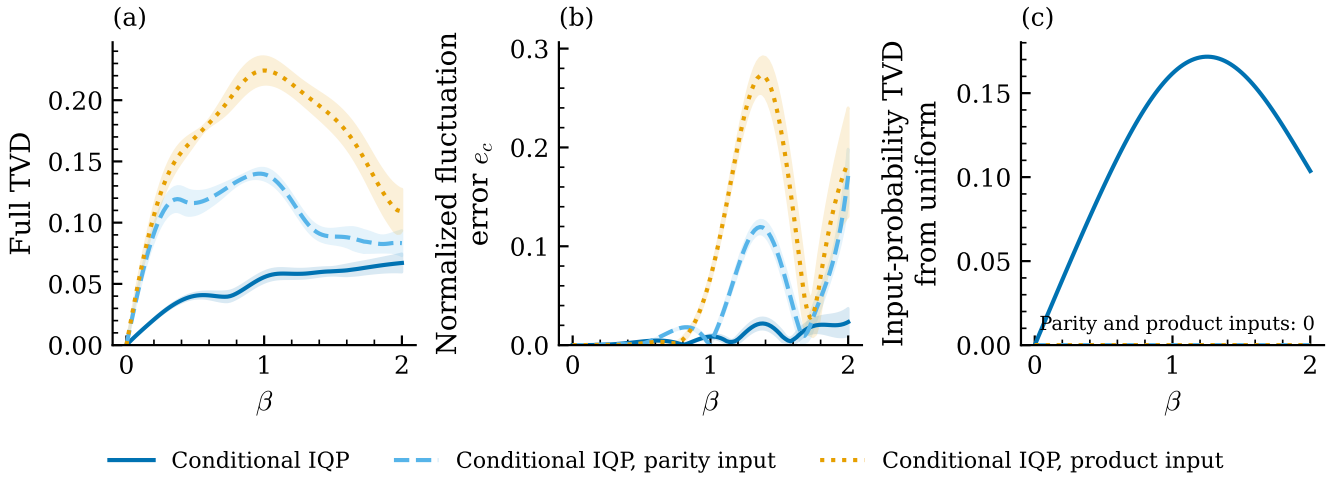


FIG. 6. Encoder comparison for conditional IQP models on the 101-point test grid. Solid curves show the squeezed parity encoder, dashed curves the parity encoder and dotted curves the product encoder. All three models use the same IQP ansatz, training temperatures and search and training budgets, with separately selected hyperparameter configurations. (a) Exact full-distribution TVD. (b) Normalized energy-fluctuation error $e_c = |c_{\mathrm{fluc},q} - c_{\mathrm{fluc},p}|/c_\star$, with $c_\star \approx 0.823$. (c) TVD between the encoder's input probabilities $|\langle x|\phi(\beta)\rangle|^2$ and the uniform distribution, which vanishes for both phase encoders. Curves and shaded bands in (a) and (b) show means and one sample standard deviation over ten independent fits. Lower values indicate smaller errors.

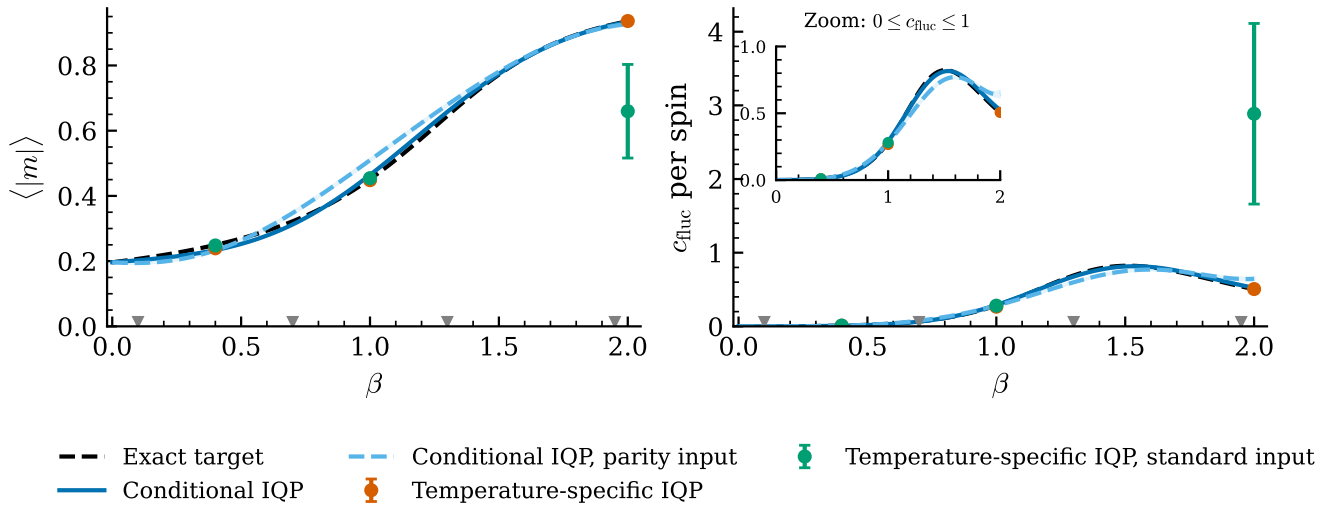


FIG. 7. Mean absolute magnetization and per-spin energy fluctuations for sixteen-qubit Curie–Weiss models. Dashed black curves show the targets. Solid and dashed curves with shading show the means and one sample standard deviation across ten independent fits for conditional IQP and the parity-input conditional model. Temperature-specific models appear as marker means with sample-standard-deviation error bars at their training temperatures, $\beta = 0.4, 1, 2$. Gray triangles mark the four conditional training temperatures. The inset enlarges the fluctuation response over $0 \le c_{\mathrm{fluc}} \le 1$, while the main panel retains the full range. We compute each observable within each fit before aggregation. Equation (61) defines the fluctuation statistic.

Figure 8 shows that the test MMD^2 decreases at every size before reaching a plateau. Over 5000 updates, the mean decreases from 0.0420 to 0.00380 at 64 qubits and from 0.0421 to 0.00710 at 2048 qubits. With the parity encoder, each update costs $O(MSPn)$ operations for M masks, S reference samples and P generators (Sec. II D 1). With the same configuration on a single GPU, peak device memory in use grew from 0.27 GB at 64 qubits to 1.8 GB at 2048 qubits. The initial phases are small, so the value at update zero approximates that of the encoder alone and provides a reference for these reductions. These results demonstrate the feasibility of

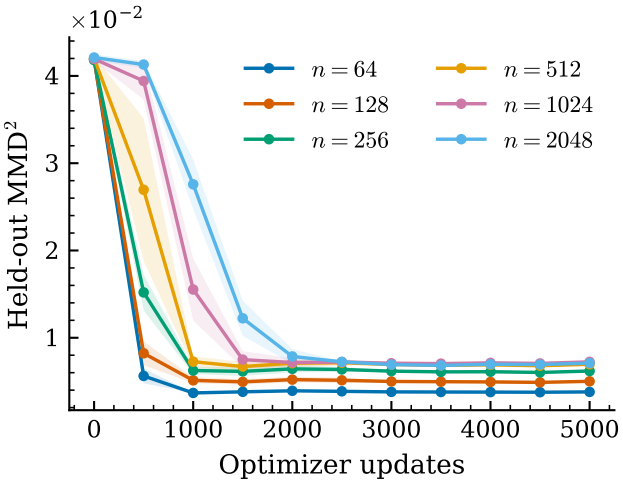


FIG. 8. Classical training of conditional IQP Born machines with 64 to 2048 qubits. All models use the parity encoder and an open rectangular lattice of trainable one- and two-body phases. At each plotted update, we classically estimate MMD^2 and average over test temperatures $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$ and two kernel bandwidths. The bandwidths give expected mask weights of two and four and vary with system size. Each fit contributes the trajectory of the initialization selected by validation at the training temperatures. Curves and shaded bands show the mean and one sample standard deviation across three fits per size.

classical conditional training up to 2048 qubits, with improvement at conditions excluded from training and selection.

The decrease in MMD^2 does not by itself establish accurate recovery of the full target distribution [48]. Although the kernels retain all correlation orders, they emphasize low-order moments, so higher-order discrepancies may remain weakly penalized or unresolved by finite sampling. The size-dependent bandwidths also limit direct comparisons of distribution accuracy across system sizes, and the sparse lattice differs from the all-to-all two-body architecture of the sixteen-qubit Curie–Weiss models.

IV. DISCUSSION

Several limitations qualify these results. First, the target families, condition ranges and encoder settings were developed through earlier exploratory runs, so the experiments assess conditional generation within these families rather than generalization to unrelated families. Second, all sixteen-qubit target families with reported accuracy are invariant under permutations of the bits, whereas the IQP architectures do not impose this symmetry, so performance on targets without permutation symmetry remains untested. Third, we test only one encoder outside the IQP-state class, the squeezed parity encoder, on one target family. Its schedule constants were fixed on development temperatures for this family, so its advantage over the parity input does not by itself establish a general rule for encoder design. The lemma of Sec. II B also shows that non-uniform input probabilities do not by themselves guarantee conditioning beyond diagonal parameters, and we do not certify how far this encoder lies from the reduced form (Appendix B 3 a). Fourth, evaluation at finitely many conditions does not establish accuracy throughout a continuous condition domain. Finally, we do not compare against the condition-specific adaptation methods discussed in the Introduction, so our results do not establish whether joint input conditioning is more accurate or efficient than fitting the condition dependence separately.

Our framework separates condition encoding from the shared unitary learned across a family of distributions. Classical estimation of the training objective and its gradients requires efficient sampling and amplitude-ratio evaluation for the reference state, rather than an encoder restricted to IQP form. This distinction allows us to use input-state design both to encode conditions and to impose exact output symmetries while retaining classical training. Unlike earlier input-conditioned QCBMs [12], the conditional model therefore remains classically trainable. The lemma of Sec. II B clarifies which encoders add to diagonal conditioning. Phase encoders cannot be absorbed into the learned parameters at the level of output states, but their reference states are IQP states, and encoders whose reference states lie outside this class remain compatible with the same training procedure.

The binomial benchmark validates learning the dependence of a target distribution on a single condition, while its two-coordinate extension tests learning a joint conditional response. Axis training exposes the model to each coordinate separately, yet the model approximates target distributions at combinations excluded from training and model selection, including outside the training convex hull. These results show that a shared IQP unitary can combine responses to separately encoded coordinates within the studied family.

The Curie–Weiss experiment shows that the choice of input state can matter as much as the learned unitary. With the same IQP ansatz, training temperatures and budgets, the squeezed parity encoder roughly halves the full-distribution error of the parity input and reduces the physical-observable score about fivefold, at held-out test temperatures. Both encoders guarantee global spin-flip symmetry for every parameter vector, so this guarantee alone does not explain the difference, and it does not ensure accurate probabilities or physical observables. Moving the reference state beyond the IQP-state class therefore provides a practical route to more accurate conditional models while keeping training classical. With the same encoder, a single shared model comes close to circuits trained separately at each test temperature. It matches their full-distribution TVD within one standard deviation at $\beta = 0.4$ and 1 but not at $\beta = 2$, and its combined observable score E_{phys} at these temperatures is

somewhat larger, 0.0172 ± 0.0045 against 0.0128 ± 0.0044. The shared model covers the whole temperature range with one circuit, whereas each temperature-specific circuit covers only its own temperature.

The same construction can enforce several prescribed output symmetries at once while retaining efficient classical reference access. The signed-graph example in Appendix D enforces six independent vertex-switching symmetries for every condition and parameter vector, without restricting the learned IQP phases.

Future work should examine how encoder design affects conditional accuracy and the costs of reference-state estimation and quantum state preparation. Our results suggest that collective operations which preserve a small symmetric description of the reference state, such as the squeezing used here, are a promising class to explore. Broader families of correlated reference states and multi-dimensional encodings could extend the tasks addressed by this framework, including targets without permutation symmetry. Matched comparisons with condition-specific adaptation would clarify when joint input conditioning is preferable. More broadly, many generative architectures reduce to sequences of condition-dependent sampling tasks. Quantum denoising diffusion models already rely on such conditioning [49–51]. Autoregressive models similarly condition each variable on those generated before it, and generation conditioned on class labels or physical parameters shares the same structure. Conditional IQP Born machines could provide classically trainable components for such models, provided that each conditioning state admits sampling-and-ratio access. Optimization also requires further study, particularly whether gradients from different conditions reinforce or oppose one another. Generator selection, initialization and adaptive kernels provide directions for investigating these questions [40–42].

Quantum deployment raises additional questions. Deploying conditional IQP circuits requires an efficient circuit for preparing $|\phi(c)\rangle$, which classical sampling-and-ratio access alone does not guarantee. The squeezing in our Curie–Weiss encoder, for example, is not a native gate and would require collective synthesis or Trotterization. Moreover, noise and circuit partitioning may affect conditional accuracy and the imposed output symmetries [52, 53]. Existing IQP sampling-hardness results apply to specific circuit families and do not establish hardness for arbitrary condition-dependent input states or specific parameter values [19, 20]. Moreover, generative models whose output distributions anticoncentrate, a property that sampling-hardness arguments typically use, are not trainable on average [54]. The binomial, two-coordinate and Curie–Weiss target families studied here admit efficient classical sampling, so these experiments do not demonstrate a quantum sampling advantage. Assessing such an advantage would require comparisons with strong classical sampling methods, together with sampling-hardness and verification arguments appropriate to the trained conditional circuits.

V. CONCLUSIONS

We introduced a conditional formulation of IQP Born machines that jointly trains a shared IQP unitary across a family of distributions. Efficient sampling and amplitude-ratio evaluation for condition-dependent reference states yield unbiased classical estimates of the conditional MMD objective and its gradients. The encoder need not have IQP form, and input-state design can enforce prescribed output bit-flip symmetries for every condition and learned parameter vector. Encoders that only apply condition-dependent phases cannot be absorbed into the learned parameters at the level of output states, but their reference states are IQP states.

At sixteen qubits, the binomial benchmark and its two-coordinate extension demonstrate conditional generation, including at joint conditions outside the training convex hull after training along each coordinate axis only. The Curie–Weiss experiment shows that a shared model can capture the temperature-dependent change from unimodal to bimodal magnetization profiles at held-out temperatures. An encoder whose reference states are not IQP states, the squeezed parity encoder, roughly halves the full-distribution error of a phase encoder under the same protocol. Training up to 2048 qubits demonstrates computational feasibility and reductions in classically estimated MMD^2 at test conditions.

These results extend IQP Born machines to conditional generation while retaining classical trainability. Full-distribution accuracy at larger scales and practical quantum sampling advantage remain open.

DATA AVAILABILITY

The code and data supporting this study are available on GitHub [45].

ACKNOWLEDGMENTS

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Appendix A: Training settings and estimator evaluation

1. Common training settings

We train all models with the unbiased squared-MMD estimator of Sec. II D 2 and keep the encoders fixed within each fit. A fit comprises one or more initializations. Each initialization draws a new parameter vector and resets the optimizer while retaining the target pools. For the binomial and scaling experiments, we draw the initial phases independently as $\theta_j \sim \mathcal{N}(0, s_\theta^2)$ and select the checkpoint with the lowest selection score across all initializations within each fit. Reported uncertainties summarize variation across independent fits.

We use `Adam` [55] with $\epsilon = 10^{-8}$, zero weight decay and cosine learning-rate decay unless stated otherwise. Table III gives the settings for the binomial and scaling experiments. Section B 3 c and Table VII give the Curie–Weiss settings, and Appendix B 2 gives the two-coordinate settings.

TABLE III. Training settings for the binomial benchmark and final scaling fits. Mask and reference budgets count draws per update. The Curie–Weiss experiments use the separate budgets in Table V. Tuned scaling values are rounded to three significant digits.

Setting	Binomial benchmark	Scaling
Updates per initialization	1,024	5,000
Initializations per fit	1	3
Independent fits	5	3
Training targets per condition	512	10,000
Target minibatch	128	450
Training mask draws	1,024	512
Training reference draws	1,024	1,024
Expected mask weights	$\{1\}$	$\{2, 4\}$
Checkpoint interval	32	100
Optimizer	`Adam`	`Adam`
Learning rate	0.005	0.00189
Learning-rate schedule	Cosine	Cosine
Initialization width s_θ	0.25	0.0130
Coefficient b_1	0.9	0.822
Final-to-initial learning-rate ratio	0.1	0.120
Coefficient b_2	0.999	0.962

For an expected mask weight w, we set the Gaussian bandwidth to

$$\sigma(n, w) = \left[-2 \ln\left(1 - \frac{2w}{n}\right)\right]^{-1/2}, \qquad 0 < w < n/2, \tag{A1}$$

where n is the number of visible qubits and w is the expected mask weight before excluding the identity mask. When a model uses several bandwidths, we weight them equally in the objective and sample one uniformly at each update. A repeated entry retains its sampling weight. We compute sixteen-qubit TVDs from exact circuit probabilities, independently of the sampled MMD estimates.

2. Efficient evaluation of the sampled estimator

For the sample values defined in Sec. II D 2, we compute the pair sums using

$$\sum_{s \neq t} Y_s Y_t = \left(\sum_s Y_s\right)^2 - \sum_s Y_s^2, \tag{A2}$$

and, since $T_b^2 = 1$,

$$\sum_{b \neq b'} T_b T_{b'} = \left(\sum_b T_b\right)^2 - B. \tag{A3}$$

For a specified phase-gradient component, the corresponding mixed pair sum satisfies

$$\sum_{s\neq t} Y_s Y'_t = \left(\sum_s Y_s\right)\left(\sum_t Y'_t\right) - \sum_s Y_s Y'_s, \tag{A4}$$

where Y'_s is the phase derivative defined in Sec. II D 2. These identities allow us to evaluate the objective in Eq. (33) and a specified gradient component in Eq. (36) using $O(S+B)$ arithmetic operations once the sample values are available.

For the sampled-mask estimator in Eq. (42), let $\mathcal{U}$ denote the set of distinct sampled masks and m_a the multiplicity of mask a. Since repeated masks use the same batches, we can write

$$\widehat{\mathrm{MMD}}^2_\sigma(c) = Q^+_\sigma \sum_{a\in\mathcal{U}} \frac{m_a}{M}\widehat{\Delta^2_c}(a). \tag{A5}$$

This regrouping preserves the sampled objective and its gradient while reducing the number of Pauli moments we evaluate.

Appendix B: Experiment protocols

1. Binomial benchmark

Each of five fits uses a fixed pool of 512 target bitstrings at each training condition $c \in \{0, 0.25, 0.5, 0.75, 1\}$. Table III gives the remaining training settings. We evaluate checkpoints at initialization, every 32 updates and at the final update, and select the checkpoint with the lowest mean exact full-distribution TVD over the development conditions $c \in \{0.125, 0.375, 0.625, 0.875\}$. The 80 test conditions are $c_j = (j + 1/3)/80$ for $j = 0, \ldots, 79$.

2. Two-coordinate benchmark

Axis training, presented in the main text, uses $(u, 0)$ at $u \in \{-1, -0.37, 0, 0.37, 1\}$ and $(0, v)$ at $v \in \{-1, -0.31, 0.31, 1\}$. Development uses $(u, 0)$ at $u \in \{-0.73, -0.17, 0.17, 0.73\}$ and $(0, v)$ at $v \in \{-0.69, -0.13, 0.13, 0.69\}$. The test set contains the sixteen mixed pairs

$$\{-1, -0.37, 0.37, 1\} \times \{-1, -0.31, 0.31, 1\}.$$

Four lie inside the training convex hull $|u| + |v| \leq 1$, and twelve lie outside. As a complementary design, grid training uses $\{-1, 0, 1\}^2$, with development conditions

$$\{-1, -0.5, 0, 0.5, 1\} \times \{-0.75, -0.25, 0.25, 0.75\}.$$

It is tested on the grid

$$u \in \{-1, -0.83, -0.37, 0.19, 0.61, 1\},$$
$$v \in \{-1, -0.79, -0.31, 0.23, 0.67, 1\},$$

excluding its four training corners, which leaves thirty-two test pairs.

Each fit uses a fixed pool of 1024 target bitstrings at each of nine training conditions. An update draws one condition and one bandwidth uniformly, 256 target strings with replacement, 1024 reference samples and 1024 nonidentity masks, with expected mask weights $\{2, 4\}$. We use `Adam` with $b_1 = 0.9$ and $b_2 = 0.999$, and cosine decay to one tenth of the initial learning rate. We select checkpoints and hyperparameters by the mean exact full-distribution TVD over the development conditions. For each design, we screen four configurations that combine initial learning rates 0.01 and 0.03 with one-body initialization means $m = 0$ and $m = \pi/3$. One-body parameters are drawn from $\mathcal{N}(m, 0.1^2)$ and pair parameters from $\mathcal{N}(0, 0.1^2)$. We confirm the two best configurations with two fits each, then perform five final fits of 2000 updates per design. Both designs select $m = \pi/3$ and learning rate 0.01. Because $m = \pi/3$ gave lower development scores, the search does not establish robustness across initialization regimes. We verified reference samples, amplitude ratios, circuit probabilities, and sampled losses and gradients against dense circuit calculations.

TABLE IV. Two-coordinate test errors. Values are means and sample standard deviations of per-fit mean errors across five independent fits. The two training designs have different test sets, so the columns do not form a paired accuracy comparison. Lower is better.

	Axis training, sixteen pairs	Grid training, thirty-two pairs
Encoder only, full and profile TVD	0.3867	0.3586
Learned model, full TVD	0.0468 ± 0.0143	0.0397 ± 0.0093
Learned model, profile TVD	0.0173 ± 0.0066	0.0156 ± 0.0036

The encoded input is invariant under qubit permutations, and the Hadamard layer preserves this symmetry. Input and reference amplitudes therefore depend only on Hamming weight. For an input bitstring of weight k, the amplitude is

$$\phi_{u,v}(k) = 2^{-8} \exp\left[\frac{iu(k-8)}{2} - \frac{iv(k-8)^2}{40}\right]. \tag{B1}$$

Grouping the terms in the Hadamard transform by input weight k and overlap ℓ gives the amplitude of any reference bitstring of weight j,

$$\psi_{u,v}(j) = 2^{-8} \sum_{k=0}^{16} \phi_{u,v}(k) \mathcal{K}_k(j), \tag{B2}$$

$$\mathcal{K}_k(j) = \sum_{\ell=0}^{k} (-1)^\ell \binom{j}{\ell} \binom{16-j}{k-\ell}, \tag{B3}$$

where ℓ counts positions occupied in both bitstrings and binomial coefficients outside their valid ranges are zero. To sample a reference bitstring, we draw its weight j with probability $\binom{16}{j}|\psi_{u,v}(j)|^2$, then draw a bitstring uniformly among those with weight j. The amplitude ratio is $R_{u,v}(z,a) = \psi_{u,v}(|z \oplus a|)/\psi_{u,v}(|z|)$, evaluated only at sampled configurations. For general n, both operations require a polynomial number of arithmetic operations.

We prepare the encoded input using sixteen Hadamard gates, sixteen $R_Z(u/2)$ rotations and 120 pairwise ZZ rotations. Since

$$J_z^2 = \frac{n}{4} I + \frac{1}{2} \sum_{i<j} Z_i Z_j,$$

applying $e^{-ivZ_iZ_j/80}$ to every pair implements the quadratic phase up to a global phase, without ancillas. We do not assess compiled depth or deployment noise.

Table IV reports both designs. Grid training gives a full-distribution TVD of 0.0397 ± 0.0093 over its thirty-two test pairs. The two designs use different test sets, so these averages are not a direct accuracy comparison. On the common 41×41 diagnostic grid of Fig. 3, which includes training and development conditions, the condition-averaged TVDs are 0.0337 for axis training and 0.0318 for grid training. This small difference gives limited evidence of an additional benefit from joint training conditions in this family.

To quantify the responses in Fig. 4, we divide each model's mean change from $u = -1$ to $u = 1$, or standard-deviation change from $v = -1$ to $v = 1$, by the corresponding exact target change. Averaged across fits, the mean-change ratios at $v = (-1, 0, 1)$ are $(0.966, 1.000, 1.043)$ for axis training and $(0.964, 0.998, 1.042)$ for grid training. The width-change ratios at $u = (-1, 0, 1)$ are $(0.917, 0.829, 0.917)$ and $(0.898, 0.814, 0.898)$, respectively.

3. Curie–Weiss models

We compare six model families for the target in Eq. (56). All use the same IQP ansatz, with sixteen one-body and 120 two-body phases and no ancillary qubits. Conditional IQP and temperature-specific IQP use the squeezed parity encoder in Eq. (57). The conditional and temperature-specific parity-input models use the parity encoder in Eq. (60). The product-input conditional model uses the product encoder in Eq. (B11), and the temperature-specific models with standard input use $|0\rangle^{\otimes 16}$.

a. Squeezed parity encoder

Every layer of the squeezed parity encoder is invariant under permutations of the fifteen leaves, so the input amplitude of a bitstring depends only on its root bit b and its leaf Hamming weight w. Let $|D_w\rangle$ denote the normalized symmetric state of the leaves with Hamming weight w. In this basis, J_z is diagonal with entries $15/2 - w$, and J_y follows from the collective ladder operators, so the squeezed leaf state

$$S(r_\beta)\,|+\rangle^{\otimes 15} = \sum_{w=0}^{15} d_w(\beta)\,|D_w\rangle \tag{B4}$$

follows from exponentiating a 16×16 matrix. Since $\sum_{i=1}^{15} Z_i Z_{16}$ takes the value $(15-2w)(1-2b)$, the input amplitude of any bitstring with root bit b and leaf weight w is

$$\phi_{\mathrm{S}}(w,b) = \frac{e^{-it_\beta(15-2w)(1-2b)/2}}{\sqrt{2}}\,\frac{d_w(\beta)}{\sqrt{\binom{15}{w}}}. \tag{B5}$$

The Hadamard layer maps a function of the leaf weight to another function of the leaf weight through the Krawtchouk transform,

$$g_b(w') = 2^{-15/2}\sum_{w=0}^{15} K_w(w')\,\phi_{\mathrm{S}}(w,b), \qquad K_w(w') = \sum_j (-1)^j \binom{w'}{j}\binom{15-w'}{w-j}, \tag{B6}$$

and the root Hadamard gives the reference amplitudes $\psi_{\mathrm{S}}(w',b') = [g_0(w') + (-1)^{b'} g_1(w')]/\sqrt{2}$. The table of 2×16 amplitudes therefore costs $O(n^3)$ operations per temperature. Global spin-flip symmetry makes the amplitudes vanish at odd total parity.

To sample the reference distribution, we draw a pair (w,b) with probability $\binom{15}{w}|\psi_{\mathrm{S}}(w,b)|^2$ and then choose a uniformly random set of w leaves. For a sampled bitstring z with leaf part z_L and a mask a with leaf part a_L, flipping changes the leaf weight to $w + |a_L| - 2\,z_L\cdot a_L$ and the root bit to $b \oplus a_{16}$, so each amplitude ratio is a quotient of two table entries and costs $O(n)$ operations. We verified the table, the sampling law and all amplitude ratios against dense constructions at small sizes.

To prepare the encoder, we apply a Hadamard to every qubit, the squeezing $S(r_\beta)$ to the leaves, and an $R_{ZZ}(t_\beta)$ rotation between each leaf and the root. The operator $J_yJ_z + J_zJ_y$ equals $\frac{1}{2}\sum_{i<j}(Y_iZ_j + Z_iY_j)$, summed over leaf pairs. Its terms do not commute, so a quantum implementation requires collective synthesis or Trotterization. Classical training uses only the amplitude table.

Section II B shows that non-uniform input probabilities do not guarantee conditioning beyond diagonal parameters. The encoder would reduce to a phase encoder with condition-dependent parameters if $U_{\boldsymbol{\lambda}}\,|\phi_{\mathrm{S}}(\beta)\rangle$ had uniform probabilities for some $\boldsymbol{\lambda}$ in the complete one- and two-body ansatz span. For the selected encoder, we minimized the TVD between these probabilities and the uniform distribution over $\boldsymbol{\lambda}$ separately at the development temperatures $\beta \in \{0.29, 0.65, 0.89, 1.25, 1.61, 1.975\}$, using six initializations at each. Without the layer, the input probabilities at these temperatures lie between 0.056 and 0.172 from uniform in TVD. The smallest residuals ϵ we found are 2×10^{-4} at $\beta = 0.29$, 0.007 at $\beta = 0.65$, between 0.035 and 0.053 for $0.89 \le \beta \le 1.61$, and 0.010 at $\beta = 1.975$. These values are upper bounds from local optimization, so they do not certify that no reducing layer exists. A residual ϵ places the input within trace distance $\sqrt{2\epsilon}$ of a phase encoder transformed by the layer, which bounds the change in every output distribution by the same amount. This bound is 0.02 at $\beta = 0.29$ and exceeds 0.25 near the crossover. The diagnostic therefore indicates that, in the hot regime, the encoder is close to a phase encoder with condition-dependent diagonal parameters, whereas near the crossover our search found no such reduction.

b. Parity encoder and training settings

The parity encoder is the binary-linear construction of Sec. II E with $Cy = (y, \bigoplus_\ell y_\ell)$ and factor amplitudes $\cos(\lambda\beta/2)$ and $-i\sin(\lambda\beta/2)$. We prepare it by applying $R_X(\lambda\beta)$ to each of the first fifteen qubits and a CNOT from each of them to qubit sixteen, followed by the Hadamard layer. For a reference bitstring (r,b), with $r \in \{0,1\}^{15}$ and $b \in \{0,1\}$, the amplitude is

$$\psi_{\mathrm{P}}(r,b) = \mathbf{1}_{b=\oplus_i r_i}\cos(\lambda\beta/2)^{15-|r|}[-i\sin(\lambda\beta/2)]^{|r|}. \tag{B7}$$

Sampling and amplitude-ratio evaluation follow Appendix C and require $O(n)$ arithmetic operations.

We fit the shared models at $\beta \in \{0.1, 0.7, 1.3, 1.95\}$. Each temperature-specific family contains three circuits fitted directly at $\beta \in \{0.4, 1, 2\}$, with separate learned phases and one jointly selected hyperparameter configuration. Each training temperature supplies a fixed pool of 1024 target bitstrings per fit. Each update draws 256 target bitstrings with replacement, 2048 nonidentity masks and 1024 reference samples. Fitting proceeds in three stages. Screening compares configurations, confirmation refits a shortlist with new fits, and final fitting produces the reported models. Table V summarizes the budgets.

TABLE V. Curie–Weiss training budgets. Initializations are counted per circuit and fit; the final column totals optimizer updates across all fits and initializations. All stages use 2000 updates per initialization, with checkpoint evaluation at initialization and every 100 updates. Each shared-model fit trains one circuit using 4096 target-pool entries. Each temperature-specific-family fit trains three circuits using 3072 entries, and therefore uses three times the learned phases and optimizer updates.

Stage	Fits	Initializations	Total updates per circuit
Screening, per proposal	5	1	10,000
Confirmation, per candidate	5	3	30,000
Final fitting, per family	10	3	60,000

c. *Hyperparameter optimization*

We select checkpoints and hyperparameter configurations using a full-distribution TVD score E_{TVD} and a physical-observable score E_{phys}. For a temperature grid Γ and a nonnegative error curve e, define

$$R_\Gamma(e) = \frac{2\max_{\beta\in\Gamma} e(\beta) + \bar{e}_\Gamma}{3}, \tag{B8}$$

where $\bar{e}_\Gamma$ is the trapezoidal integral over the grid divided by its temperature span. This score gives the maximum error twice the weight of the temperature-averaged error. For a single temperature, both terms equal the error at that temperature. We compute $E_{\mathrm{TVD}} = R_\Gamma(\mathrm{TVD})$ from exact probabilities over all 2^{16} bitstrings. The physical-observable score is

$$E_{\mathrm{phys}} = \frac{R_\Gamma(e_m) + R_\Gamma(e_c)}{2}, \qquad e_m = \left|\langle|m|\rangle_q - \langle|m|\rangle_p\right|, \tag{B9}$$

with $e_c = |c_{\mathrm{fluc},q} - c_{\mathrm{fluc},p}|/c_\star$, where $c_\star \approx 0.823$ is the maximum target value of c_{fluc} for $0 \le \beta \le 2$. Appendix B 3 f defines these observables.

For shared models, checkpoint selection uses the development grid

$$\begin{aligned}\Gamma_{\mathrm{dev}} = &\{0.05 + 0.12j : j = 0, \ldots, 15\} \\ &\cup \{1.975\}.\end{aligned} \tag{B10}$$

Within each fit, we pool all scheduled checkpoints across initializations and find the minimum score $E^{\min}_{\mathrm{TVD}}$. We then select the checkpoint with the smallest E_{phys} among those satisfying $E_{\mathrm{TVD}} \le E^{\min}_{\mathrm{TVD}} + 0.01$. Temperature-specific models use the same rule at their training temperatures, combining the errors of a family's three circuits through R_Γ.

We run one `Optuna` study per family [56], with 60 random configurations followed by 60 proposals from a tree-structured Parzen estimator that receives only E_{TVD}. Table VI specifies the search space. The cosine schedule decays to one tenth of the initial learning rate over 2000 updates, and the encoder-scale range gives $\lambda\beta \le 3 < \pi$ over $\beta \in [0, 2]$. For the squeezed parity encoder, we search the schedule constant a in Eq. (59) instead of λ, because it has different units, and we fix $\kappa = 0.4$. All other ranges and priors are identical across families, and all families share the random streams for target pools and stochastic estimates at each stage, fit and initialization. No screening proposal or fit failed numerically.

Confirmation refits three or four leading configurations per family with five new fits each. For shared models, it ranks configurations on Γ_{dev} together with a separate grid of twenty interleaved temperatures, while still selecting checkpoints on Γ_{dev}. We select the configuration with the smallest confirmed E_{phys} among those within 0.01 of the minimum confirmed E_{TVD}, then perform ten final fits per family with three initializations per circuit. Table VII reports the selected configurations.

The shared-model test grid contains 101 points: 0, 2 and $(6j-4)/297$ for $j = 1, \ldots, 99$. We also evaluate the shared models at $\beta \in \{0.4, 1, 2\}$. Neither the test grid nor these temperatures enter shared-model training or selection.

TABLE VI. Curie–Weiss hyperparameter search space and initial random sampling distributions. TPE adapts its proposals after the random stage. We use `Adam` for all models. The encoder scale λ applies to the parity and product encoders, and the schedule constant a to the squeezed parity encoder; the standard-input family has no encoder-scale parameter.

Parameter	Options or range	Initial prior
Learning rate	$[3\times10^{-4}, 10^{-1}]$	Log-uniform
Width s_θ	$[0.01, 1]$ radians	Log-uniform
b_1	$[0.8, 0.95]$	Uniform
$1-b_2$	$[0.001, 0.05]$	Log-uniform
Schedule	Constant, cosine	Equal probabilities
Encoder scale λ	$[0.25, 1.5]$	Log-uniform
Schedule constant a	$[0.3, 0.7]$	Log-uniform
Kernel pair (w_2, w_3)	45 pairs, $w_2 \le w_3$	Uniform categorical

The expected mask-weight entries are $(2, w_2, w_3)$, where $w_2 \le w_3$ and $w_2, w_3 \in \{4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 7.99\}$. Repeated entries are allowed and retain their sampling weight. The upper value 7.99 keeps $w < n/2$.

TABLE VII. Hyperparameter configurations selected by confirmation. The encoder scale is a for the squeezed parity encoder, whose squeezing strength is fixed at $\kappa = 0.4$, and λ for the parity and product encoders. Each temperature-specific family's settings apply jointly to three circuits with separate learned phases. Widths are in radians, and b_1 and b_2 are the `Adam` moment parameters. Continuous values are rounded to three significant digits.

Model	Learning rate	Width s_θ	b_1	b_2	Schedule	Encoder scale	Kernel weights
Conditional IQP	0.00354	0.0341	0.829	0.998	constant	0.492	[2, 4, 7.5]
Conditional IQP, parity input	0.00342	0.0221	0.802	0.992	cosine	0.684	[2, 4, 7.5]
Conditional IQP, product input	0.0301	0.0455	0.826	0.999	constant	0.657	[2, 4, 5]
Temperature-specific IQP	0.000724	0.0517	0.803	0.967	constant	0.471	[2, 5, 5]
Temperature-specific IQP, parity input	0.00430	0.0300	0.899	0.966	constant	0.582	[2, 7.5, 7.99]
Temperature-specific IQP, standard input	0.0898	0.0326	0.946	0.999	cosine	–	[2, 5, 5.5]

d. Encoder comparison

The product encoder is

$$|\phi_{\mathrm{prod}}(\beta;\lambda)\rangle = \bigotimes_{i=1}^{16} R_Z(\lambda\beta) R_Y(\pi/2)\,|0\rangle_i\,. \tag{B11}$$

Like the parity encoder, it gives uniform computational-basis input probabilities, but it does not enforce global spin-flip symmetry. The three conditional models in Fig. 6 use the same 136 IQP phases, training temperatures, search ranges and budgets per circuit, with separate hyperparameter studies.

Table VIII reports the aggregate scores. The temperature-averaged TVD is 0.0474 ± 0.0028 for the squeezed parity encoder, 0.1033 ± 0.0036 for the parity encoder and 0.1662 ± 0.0063 for the product encoder, with per-fit maximum TVDs of 0.0677 ± 0.0070, 0.1406 ± 0.0037 and 0.2245 ± 0.0112. Relative to the product input, the parity input reduces E_{TVD} by 37.5% and E_{phys} by 37.9%. Relative to the parity input, the squeezed parity input reduces them by 52.5% and 79.1%. The fluctuation-error curves of the two phase encoders cross, so a lower aggregate E_{phys} does not imply lower error for every observable at every temperature.

To separate the squeezing from the angle schedule, we refit the selected squeezed configuration with $\kappa = 0$. This removes the squeezing but keeps the schedule $\sin t_\beta = \tanh(a\beta^2)$, so the encoder becomes a phase encoder. We follow the final-stage protocol, with ten fits paired with the reported fits, at the selected $a = 0.492$ and at the ends of its search range, $a = 0.3$ and 0.7. The optimizer settings remain those selected for $\kappa = 0.4$, without a new search. The resulting E_{TVD} is 0.1692 ± 0.0073, 0.4296 ± 0.0067 and 0.1878 ± 0.0197 at $a = 0.492$, 0.3 and 0.7, and the temperature-averaged TVD is 0.1237 ± 0.0029, 0.3001 ± 0.0047 and 0.1491 ± 0.0095. Every fit has a higher E_{TVD} than its paired parity-input fit. Within this range of a, the schedule alone therefore does not reproduce the improvement, which requires the squeezing.

TABLE VIII. Conditional IQP models with different encoders and the same IQP ansatz, training temperatures and budgets. The three temperature columns report exact full-distribution TVD. The aggregate scores E_{TVD} and E_{phys}, defined in the hyperparameter-optimization protocol, use the 101-point test grid. Values are means ± sample standard deviations over ten independent fits. Lower values indicate smaller errors.

Model	$\beta = 0.4$	$\beta = 1$	$\beta = 2$	E_{TVD}	E_{phys}
Conditional IQP	0.0366 ± 0.0016	0.0555 ± 0.0043	0.0671 ± 0.0075	0.0609 ± 0.0055	0.0181 ± 0.0030
Conditional IQP, parity input	0.1185 ± 0.0107	0.1396 ± 0.0048	0.0832 ± 0.0103	0.1281 ± 0.0027	0.0865 ± 0.0088
Conditional IQP, product input	0.1576 ± 0.0079	0.2242 ± 0.0113	0.1104 ± 0.0169	0.2051 ± 0.0088	0.1393 ± 0.0107

TABLE IX. Exact Hamming-weight profile TVD at sixteen qubits. Values are means ± sample standard deviations over ten independent fits, computed within each fit before aggregation. Lower is better. Conditional and temperature-specific IQP use the squeezed parity encoder; the parity-input models use the parity encoder, and the standard-input model uses $|0\rangle^{\otimes 16}$. Conditional models exclude these test temperatures from training and selection, whereas temperature-specific models are trained directly at each temperature.

Model	$\beta = 0.4$	$\beta = 1$	$\beta = 2$
Conditional IQP	0.0347 ± 0.0018	0.0232 ± 0.0018	0.0388 ± 0.0102
Conditional IQP, parity input	0.0438 ± 0.0040	0.1203 ± 0.0046	0.0646 ± 0.0124
Temperature-specific IQP	0.0205 ± 0.0043	0.0193 ± 0.0067	0.0248 ± 0.0094
Temperature-specific IQP, parity input	0.0149 ± 0.0065	0.0642 ± 0.0028	0.1298 ± 0.0065
Temperature-specific IQP, standard input	0.0281 ± 0.0247	0.0833 ± 0.0158	0.4003 ± 0.1736

e. *Profile and full-distribution errors*

Table IX reports profile TVD from the same ten fits, computed within each fit before averaging. Profile and full-distribution errors can rank models differently. At $\beta = 1$, the parity-input temperature-specific circuits have smaller profile TVD than the parity-input conditional model, 0.0642 ± 0.0028 versus 0.1203 ± 0.0046, although their full TVD is larger. Conditional IQP has profile TVD 0.0232 ± 0.0018 at $\beta = 1$. At $\beta = 0.4$, its profile TVD of 0.0347 ± 0.0018 accounts for most of its full TVD of 0.0366 ± 0.0016, so its remaining hot-regime error lies mainly in the magnetization profile rather than within Hamming-weight shells. The standard-input model has profile TVD 0.0281 ± 0.0247 but full TVD 0.3787 ± 0.0034 at $\beta = 0.4$. Temperature-specific IQP has lower profile TVD than conditional IQP at all three temperatures, 0.0205 ± 0.0043, 0.0193 ± 0.0067 and 0.0248 ± 0.0094.

f. *Physical observables*

For a distribution q, let $\rho_q(k \mid \beta)$ denote its exact Hamming-weight profile and set $m_k = (2k - n)/n$ with $n = 16$. We compute its moments as $\langle m^r \rangle_q = \sum_{k=0}^{n} \rho_q(k \mid \beta) m_k^r$ and its mean absolute magnetization as $\sum_k \rho_q(k \mid \beta)|m_k|$. With $H = -nm^2/2$ in units $J = k_B = 1$, Eq. (61) becomes

$$c_{\mathrm{fluc},q}(\beta) = \frac{\beta^2 n}{4} \left(\langle m^4 \rangle_q - \langle m^2 \rangle_q^2 \right). \tag{B12}$$

These observables depend only on the Hamming-weight profile. For the Gibbs target, this expression equals the heat capacity per spin, but we do not assume that the learned family satisfies the same thermodynamic identity. We compute each observable within each fit before averaging.

Table X reports e_m and e_c at the test temperatures. On these three temperatures, E_{phys} is 0.0172 ± 0.0045 for conditional IQP, 0.0914 ± 0.0102 and 0.1130 ± 0.0223 for the parity-input and product-input conditional models, 0.0128 ± 0.0044 for temperature-specific IQP, 0.0363 ± 0.0018 for the parity-input temperature-specific circuits and 1.2223 ± 0.6289 for the standard-input circuits. The last value reflects substantial variation across fits at $\beta = 2$, and we retain all ten fits.

4. Scaling

We use the parity encoder in Eq. (60), generalized to n qubits, with selected scale $\lambda = 1.18$, and the open rectangular lattices in Table XI. At size n, the target is $p(x \mid \beta) \propto \exp[\beta(2|x| - n)^2/(2n)]$. We train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ with

TABLE X. Physical-observable errors for sixteen-qubit Curie–Weiss models at three test temperatures. The mean absolute magnetization error is $e_m = |\langle|m|\rangle_q - \langle|m|\rangle_p|$, and the normalized energy-fluctuation error is $e_c = |c_{\mathrm{fluc},q} - c_{\mathrm{fluc},p}|/c_\star$, with $c_\star \approx 0.823$ the maximum target value for $0 \le \beta \le 2$. Values are means ± sample standard deviations over ten independent fits, with errors computed within each fit before averaging. Lower values indicate smaller errors. Conditional models exclude these temperatures from training and selection, whereas temperature-specific models are trained directly at each temperature.

Model	Error	$\beta = 0.4$	$\beta = 1$	$\beta = 2$
Conditional IQP	e_m	0.0159 ± 0.0008	0.0129 ± 0.0013	0.0023 ± 0.0013
Conditional IQP	e_c	0.0021 ± 0.0001	0.0087 ± 0.0018	0.0234 ± 0.0138
Conditional IQP, parity input	e_m	0.0167 ± 0.0017	0.0581 ± 0.0028	0.0083 ± 0.0043
Conditional IQP, parity input	e_c	0.0009 ± 0.0002	0.0028 ± 0.0015	0.1712 ± 0.0272
Conditional IQP, product input	e_m	0.0174 ± 0.0039	0.0834 ± 0.0103	0.0130 ± 0.0066
Conditional IQP, product input	e_c	0.0005 ± 0.0004	0.0683 ± 0.0034	0.1851 ± 0.0551
Temperature-specific IQP	e_m	0.0104 ± 0.0020	0.0035 ± 0.0039	0.0014 ± 0.0013
Temperature-specific IQP	e_c	0.0020 ± 0.0003	0.0196 ± 0.0080	0.0057 ± 0.0084
Temperature-specific IQP, parity input	e_m	0.0006 ± 0.0004	0.0023 ± 0.0018	0.0127 ± 0.0013
Temperature-specific IQP, parity input	e_c	0.0013 ± 0.0003	0.0475 ± 0.0042	0.0705 ± 0.0053
Temperature-specific IQP, standard input	e_m	0.0022 ± 0.0028	0.0030 ± 0.0024	0.2759 ± 0.1435
Temperature-specific IQP, standard input	e_c	0.0003 ± 0.0002	0.0036 ± 0.0023	2.8935 ± 1.4896

fixed pools of 10,000 target samples per condition, and draw separate validation pools of 2000 targets per training condition. Table III gives the final training settings.

TABLE XI. Open rectangular lattice architectures for the scaling experiments. We assign one trainable phase to each qubit and one to each horizontal or vertical nearest-neighbor pair.

Visible qubits	Lattice	Trainable parameters
64	8×8	176
128	8×16	360
256	16×16	736
512	16×32	1,488
1,024	32×32	3,008
2,048	32×64	6,048

We evaluate checkpoints at initialization and every 100 updates, using four fixed validation sets. Each set contains 512 nonidentity mask draws per condition and bandwidth and 1024 reference draws per condition. Let $\overline{M}_n^2(\boldsymbol{\theta}; \beta)$ denote the validation MMD^2 estimate averaged over the four sets and two bandwidths. We select checkpoints using

$$V_n(\boldsymbol{\theta}) = \frac{1}{4} \sum_{\beta \in \{0.6, 0.8, 1.2, 1.4\}} \frac{\overline{M}_n^2(\boldsymbol{\theta}; \beta)}{d_{n,\beta}}. \tag{B13}$$

Each fixed scale $d_{n,\beta}$ is the median training-temperature MMD^2 across three earlier baseline fits, floored at 10^{-3}, which balances the temperatures' contributions to selection. We retain negative unbiased estimates.

We optimize the hyperparameters in Table XII at 64 and 256 qubits. Screening evaluates 48 proposals, comprising a baseline configuration, a proposal from earlier sixteen-qubit parity-model optimization, 14 random proposals and 32 tree-structured Parzen estimator proposals, each with one 2000-update initialization per size. We confirm the

TABLE XII. Scaling hyperparameter search space and initial random sampling distributions.

Hyperparameter	Range	Initial random sampling
Initial learning rate	$[10^{-4}, 0.05]$	Log-uniform
Initialization width s_θ (radians)	$[0.005, 0.3]$	Log-uniform
Encoder scale λ	$[0.25, 1.5]$	Uniform
b_1	$[0.8, 0.95]$	Uniform
b_2	$[0.95, 0.999]$	Uniform
Final-to-initial learning-rate ratio	$[0.05, 1]$	Log-uniform

three leading proposals and the baseline with three 5000-update fits per size. The lowest mean confirmation score determines the configuration used at all six sizes.

For evaluation, we draw five test sets independently of training and selection. Each contains 2000 target draws per test condition, 512 mask draws per condition and bandwidth, and 1024 reference draws per condition. At update 0 and every 500 updates, we average the estimated MMD^2 over the five sets, test temperatures and two bandwidths to obtain one value per fit. These values therefore combine variation from training and sampled evaluation.

Appendix C: Binary-linear conditional encoders

This appendix establishes normalization, efficient classical access and state preparation for the binary-linear encoder of Sec. II E. We take a full-column-rank matrix $C \in \mathbb{F}_2^{n\times k}$ and normalized factor amplitudes $f_\ell(0;c)$ and $f_\ell(1;c)$, with all matrix operations over $\mathbb{F}_2$.

1. Normalization and sampling

Since C is injective, distinct latent strings y give orthogonal computational-basis states $|Cy\rangle$. The reference-state norm therefore satisfies

$$\begin{aligned}\langle\psi_c|\psi_c\rangle &= \sum_{y\in\mathbb{F}_2^k}\prod_{\ell=1}^{k}|f_\ell(y_\ell;c)|^2 \\ &= \prod_{\ell=1}^{k}\left(|f_\ell(0;c)|^2+|f_\ell(1;c)|^2\right)=1.\end{aligned} \tag{C1}$$

The probability of a reference configuration Cy is $\prod_\ell |f_\ell(y_\ell;c)|^2$. We can therefore sample the reference state by drawing the latent bits independently and returning Cy.

2. Amplitude ratios

We precompute a left inverse $L \in \mathbb{F}_2^{k\times n}$ satisfying $LC = I_k$. Given a reference sample z, we recover its latent coordinates as $y = Lz$. For a mask a, we compute $d = La$ and test whether $Cd = a$. If the test fails, $a \notin \operatorname{im} C$, so $z\oplus a$ lies outside the reference support and the amplitude ratio is zero. If the test succeeds, $z\oplus a = C(y\oplus d)$, and Eq. (45) gives the ratio as a product of factor ratios. Sampled configurations have nonzero amplitude, so the denominators are nonzero. After computing L once by Gaussian elimination, reference sampling and ratio evaluation each require $O(nk)$ elementary operations, assuming bounded cost for evaluating the factors and their ratios.

3. State preparation

We extend the columns of C to an invertible matrix $M \in \mathbb{F}_2^{n\times n}$. We prepare the first k qubits in the factor states

$$|f_\ell(c)\rangle = f_\ell(0;c)\,|0\rangle + f_\ell(1;c)\,|1\rangle\,, \tag{C2}$$

and initialize the remaining $n-k$ qubits in $|0\rangle$. A reversible linear circuit implementing M maps each basis state $|y,0^{n-k}\rangle$ to $|Cy\rangle$ and therefore maps the product of the factor states and zero states to $|\psi_c\rangle$. Standard binary elimination implements this map using $O(n^2)$ CNOT gates with unrestricted connectivity and no ancillary qubits [57]. We then apply $H^{\otimes n}$ to prepare $|\phi(c)\rangle$. The construction is efficient provided that the rotation angles of the factor states can be computed efficiently from c.

Appendix D: Signed-graph encoder with exact switching symmetry

This appendix shows how the binary-linear construction of Sec. II E enforces several independent bit-flip symmetries at once. Sixteen qubits represent the edge signs $s_{ij} = (-1)^{x_{ij}}$ of a seven-vertex graph in which every pair among vertices $0,\dots,4$ is connected and each of vertices 5 and 6 is connected to vertices $0,1,2$ [Fig. 9(a)]. We consider the Gibbs family

$$p_\beta(x) = Z_\beta^{-1}e^{\beta S(x)}, \qquad S(x) = \sum_{\{i,j,k\}\in\mathcal{T}} s_{ij}s_{ik}s_{jk}, \tag{D1}$$

where $\mathcal{T}$ contains the graph's sixteen triangles, so increasing β favors balanced triangles. A vertex switch reverses the signs of all edges incident on a vertex. Every triangle contains zero or two of these edges, so switches preserve p_β.

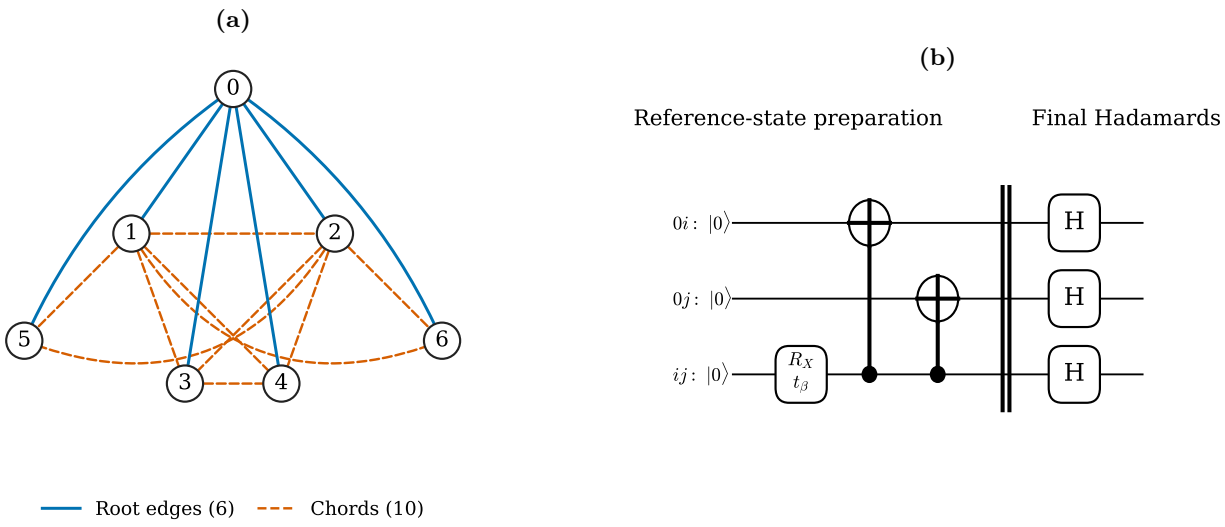


FIG. 9. Signed-graph encoder with six independent switching symmetries. (a) The seven-vertex graph has sixteen edges. Six root edges form a spanning tree, and each of the ten remaining edges forms a fundamental triangle with two root edges. (b) Reference-state preparation on an isolated triangle. A chord rotation $R_X(t_\beta)$ and two CNOTs prepare $\cos(t_\beta/2)\,|000\rangle - i\sin(t_\beta/2)\,|111\rangle$ before the displayed Hadamards. For the full encoder, we repeat the rotation–CNOT construction over all ten chords, combining contributions to shared root edges modulo two, then apply one global Hadamard layer.

Fixing vertex 0 removes the redundant global switch, and the six remaining switches generate a group of 64 edge-bit masks, which we denote by $\mathcal{S}$.

The six root edges $0i$ form a spanning tree, and each of the ten remaining edges, the chords $E_{\rm ch}$, closes a triangle with two root edges. These ten triangles form a cycle basis and supply the columns of C. The encoded input is

$$|\phi_\beta\rangle = \exp\left[-\frac{it_\beta}{2}\sum_{(i,j)\in E_{\rm ch}} Z_{0i}Z_{0j}Z_{ij}\right]|+\rangle^{\otimes 16}, \tag{D2}$$

$$t_\beta = \frac{\pi}{2}\tanh(1.5\beta), \tag{D3}$$

which is a phase encoder. Every supported reference configuration has even parity on the edges incident on each vertex, so $X_h\,|\phi_\beta\rangle = |\phi_\beta\rangle$ for every $h \in \mathcal{S}$. Equation (46) then gives $q_{\boldsymbol{\theta}}(x \mid \beta) = q_{\boldsymbol{\theta}}(x \oplus h \mid \beta)$ for every condition and parameter vector. This guarantee does not depend on the temperature map and requires no training to verify, so the example concerns encoder design and classical access rather than learning accuracy.

1. Cycle basis and reference state

We now give the matrices and access procedures, using column vectors over $\mathbb{F}_2$. We order the sixteen edges lexicographically as

$$(01, 02, 03, 04, 05, 06, 12, 13, 14, 15, 16, 23, 24, 25, 26, 34). \tag{D4}$$

The root edges $01, \ldots, 06$ form a spanning tree, and the chords are $E_{\rm ch} = \{12, 13, 14, 15, 16, 23, 24, 25, 26, 34\}$. Each column of $C \in \mathbb{F}_2^{16\times 10}$ contains ones on the three edges of one fundamental triangle, with columns ordered as the chords in $E_{\rm ch}$. The chord rows therefore form the identity matrix, so C has rank ten. Since the connected graph has cycle rank $16 - 7 + 1 = 10$, its columns form a basis of the cycle space.

For chord bits $y \in \mathbb{F}_2^{10}$, the reference state is

$$|\psi_\beta\rangle = \sum_{y\in\mathbb{F}_2^{10}} \alpha_\beta(y)\,|Cy\rangle\,, \tag{D5}$$

$$\alpha_\beta(y) = \cos(t_\beta/2)^{10-|y|}\,[-i\sin(t_\beta/2)]^{|y|}, \tag{D6}$$

with $t_\beta = (\pi/2)\tanh(1.5\beta)$ and encoded input $|\phi_\beta\rangle = H^{\otimes 16}|\psi_\beta\rangle$. Appendix C establishes its normalization.

Each vertex switch selects either zero or two edges of every triangle, so $C^{\mathsf{T}}h = 0$ for every $h \in \mathcal{S}$. Since $\ker C^{\mathsf{T}}$ has dimension $16-10=6$ and therefore contains 64 masks, $\mathcal{S} = \ker C^{\mathsf{T}}$. Equation (46) then gives $q_{\boldsymbol{\theta}}(x \mid \beta) = q_{\boldsymbol{\theta}}(x \oplus h \mid \beta)$ for every $h \in \mathcal{S}$, condition and parameter vector.

2. Preparation circuit

We prepare the reference state by initializing all sixteen qubits in $|0\rangle$. For each chord ij, we apply $R_X(t_\beta)$ to its qubit and two CNOT gates controlled by that qubit, targeting root edges $0i$ and $0j$. On an isolated triangle, these operations prepare $\cos(t_\beta/2)|000\rangle - i\sin(t_\beta/2)|111\rangle$, as illustrated in Fig. 9(b). In the full graph, the CNOTs combine contributions to shared root edges modulo two. A final Hadamard layer gives the encoded input. The preparation uses ten rotations, twenty CNOTs and sixteen Hadamards, without ancillary qubits.

3. Classical sampling and amplitude ratios

We sample the reference distribution by drawing the ten chord bits independently, each equal to one with probability $\sin^2(t_\beta/2)$, and returning Cy. Because the chord rows of C form the identity, the left inverse L of Appendix C reads off the chord entries, so a sample's chord bits equal y and a mask's chord bits give d. If $Cd \neq a$, the mask lies outside the cycle space and the ratio is zero. Otherwise, the flip maps Cy to $C(y \oplus d)$, and for $\beta \neq 0$

$$R_\beta(Cy, a) = \left[-i\tan\left(\frac{t_\beta}{2}\right)\right]^{|y\oplus d|-|y|}. \tag{D7}$$

At $\beta = 0$, the reference state is $|0\rangle^{\otimes 16}$, so the ratio equals one for $a = 0$ and zero otherwise.